ELECTRIC CURRENTS, RANDOM WALKS AND ANALYSIS ON GRAPHS

JOHN E. HUTCHINSON

Abstract. We build up the basics of finite graphs and their interpretations via electric currents, random walks and matrices, sufficient to develop analysis on fractals.

Contents

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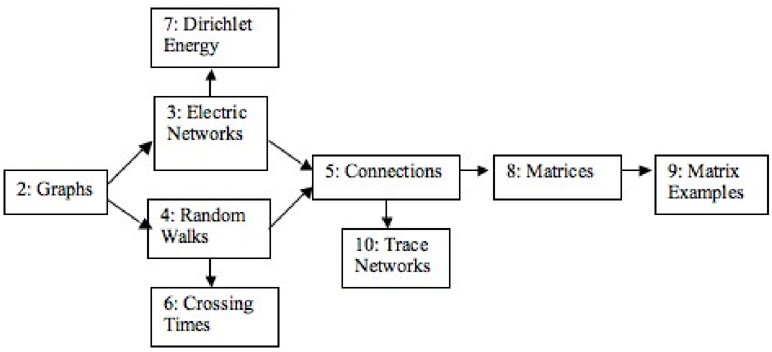

Figure 1. Section dependencies, where $A \to B$ means A is a prerequisite for B. The prerequisite relation is transitive.

*** (Needs more info)*** Section dependencies are indicated in Figure 1. Sometimes there are forward references for alternative proofs, examples, etc., but this is minor and not essential to the development.

1. **Introduction**

Section:

2 We introduce the idea of a graph with weights on it edges.

3 We show how the graph can be interpreted as an electrical network, where the weight on each edge is the conductance, i.e. the reciprocal of the resistance, of the corresponding wire. The current between connected nodes is given by the voltage difference and the conductance, according to Ohm's law. Induced voltages are then given by solving Laplace's equation, and this is essentially Kirchoff's law. The normalised current (normalised by total conductance at wires connected to that node) out of a node is the laplacian of the voltage at that node.

We introduce the idea of the effective resistance $\mathbb{R}_{ab}$ (the reciprocal is the effective conductance) between two nodes a and b. We see this corresponds to solving Laplace's equation with $v(b) = 0$ and either a pointwise or Laplacian condition at a. The first is a Dirichlet problem and the second is finding the Green's function — in the discrete setting as here the two are equivalent and this gives useful information.

We show $\mathbb{R}_{ab}$ defines a metric.

4 Etc., etc.

2. **Graphs**

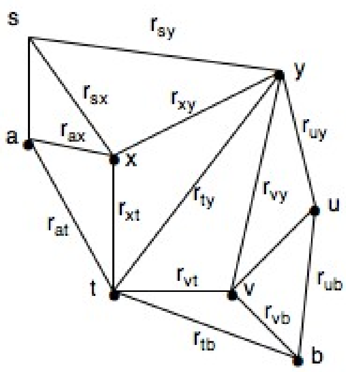

FIGURE 2. A finite graph

A *(finite) graph* is a triple

$$\mathbb{G} = (V, E, r_{xy})$$

where

- V is the finite set of *vertices* or *nodes*,
- E is the set of *edges* and consists of certain pairs of distinct vertices,
- r_{xy} is a (strictly) *positive* function defined on E.

Note $r_{xy} = r_{yx}$.

We assume the graph to be *connected*.

2.1. **Notation and Definitions.**

- Write $x \sim y$ if $\{x, y\} \in E$.
- A *simple* graph is one such that $r_{xy} \equiv 1$ for $\{x, y\} \in E$.
- $c_{xy} := 1/r_{xy}$.

  If xy is not an edge define $c_{xy} = 0$.

  r_{xy} is the *resistance* of the edge xy and c_{xy} is the *conductance*.
- The *weighted degree* of x is $c_x := \sum_y c_{xy}$. It is the number of edges from x if $\mathbb{G}$ is simple and in this case is just called the *degree* of x.
- The *total mass* of $\mathbb{G}$ is $C = C(\mathbb{G}) := \sum_x c_x$. In the case of a simple graph it is twice the number of edges, or equivalently the sum of the degrees of the vertices.

3. **Electric Network Interpretation of a Graph**

3.1. **Physical Motivation.** Fix the graph $\mathbb{G}$. Think of $\mathbb{G}$ as an electric network [EN] with resistance r_{xy} on each edge xy. Suppose $\emptyset \neq G \subset V$ and $a \in V \setminus G$. Voltages $v(a) \neq 0$, and $v(b) = 0$ for $b \in G$, are imposed. (That is, the circuit is grounded on G.)

An important case is $G = \{b\}$ contains a single node. The set of nodes $\{a\} \cup G$ is the set of *boundary* nodes. Later we impose more general boundary conditions on v. The *interior* nodes are $V^\circ = V \setminus G$.

- Each node x settles into a steady state voltage $v(x)$.
- A *current* i_{xy} flows from x to y where
$$i_{xy} = (v_x - v_y)c_{xy}. \quad \textit{(Ohm's law)}$$
Thus $i_{xy} = -i_{yx}$ and current flows "downhill".
- For x an interior node there is *zero net current* flowing out of x, i.e. into $\mathbb{G}$ through x:
$$\sum_y i_{xy} \left(\text{i.e. } \sum_y (v_x - v_y)c_{xy}\right) = 0. \quad \textit{(Kirchoff's law)}$$

3.2. **Mathematical Framework.** Because of the previous physical motivation[1] define the *voltage* to be the unique (see below) function v with[2]

$$\text{prescribed } v(a) > 0, \quad v(b) = 0 \text{ for } b \in G, \tag{1}$$

$$v(x) = \frac{1}{c_x}\sum_{y\sim x} c_{xy}v(y), \quad x \neq a, x \notin G. \tag{2}$$

Thus for interior x, $v(x)$ is a *weighted average* of v at neighbouring nodes.

Note that the second set of boundary conditions and (2) are *homogeneous*, i.e. preserved under addition and scalar multiplication of functions.

The following are true for general boundary voltages, with the same proofs.

- Denote the *Laplacian* of $f : V \to \mathbb{R}$ by[3]
$$\boxed{\triangle f(x) := \left(\frac{1}{c_x}\sum_{y\sim x} c_{xy}f(y)\right) - f(x) = \frac{1}{c_x}\sum_{y\sim x} c_{xy}\big(f(y) - f(x)\big).} \tag{3}$$
Thus (2) implies
$$\triangle v(x) = 0, \quad x \neq a, x \notin G. \tag{4}$$
For this reason we say v is *harmonic* at interior nodes.
- *Maximum Principle*: For interior x; $v(x) \leq \max_{y\sim x} v(y)$ by (2). Thus
$$0 \leq v(x) \leq v(a) \text{ for all } x. \tag{5}$$
- *Uniqueness*: Any two solutions of (1) and (2) are equal. (Apply (5) to the difference of the solutions.)
- *Existence*: There *exists* a (unique) solution of (2). (Since uniqueness implies existence for n linear equations in n variables.)
- The *current* along the (directed) edge xy is $i_{xy} := (v_x - v_y)c_{xy}$. Clearly $i_{xy} = -i_{yx}$.
- The *current out of* $x \in V$, or *into* $\mathbb{G}$ *at* x, is defined by
$$\boxed{i_x := \sum_y i_{xy} = -c_x \triangle v(x).} \tag{6}$$
Thinking of the case $i_x < 0$, $-i_x$ is the current *out of* $\mathbb{G}$ *at* x.

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[1] Equation (2) is a rewriting of Kirchoff's law at x.

[2] We could as well take $v(a) < 0$, but it is convenient to take a positive voltage at a as we then have a current flowing from a *into* $\mathbb{G}$ and then *out of* $\mathbb{G}$ through G.

[3] So $\triangle f(a)$ is the difference between the weighted average of f at neighbours of a and the value $f(a)$ itself.

Note $i_x = 0$ at interior nodes by (4). The *flux* of i through the network is[4]

$$F(i) := i_a = -\sum_{b\in G} i_b. \tag{7}$$

3.3. **Rescaling Solutions.** We make a simple but very useful observation which depends essentially on the homogeneity in (1) and (2).

If f_1 and f_2 are the solutions of (1) + (2) with prescribed voltages at a given by $f_1(a)$ and $f_2(a)$ respectively, then trivially by uniqueness and homogeneity, f_1 and f_2 are equal up to rescaling:

$$f_1(x) = \frac{f_1(a)}{f_2(a)} f_2(x), \ \forall x. \tag{8}$$

Moreover,

$$\frac{f_1(a)}{f_2(a)} = \frac{\Delta f_1(a)}{\Delta f_2(a)}. \tag{9}$$

So either of the above can be used as the scaling factor in (8).

3.4. **Effective Conductance and Resistance.** The important case is $G = \{b\}$.

- Motivated by Ohm's law (see Section 3.1) the *effective conductance* and *effective resistance* between a and G are defined by

$$\mathbb{C}_{aG} = \frac{1}{\mathbb{R}_{aG}} = \frac{i_a}{v(a)}. \tag{10}$$

  Note $\mathbb{C}_{aG}$ and $\mathbb{R}_{aG}$ are independent of $v(a)$ by (8).
- *Symmetry:*[5]

$$\mathbb{C}_{ab} = \mathbb{C}_{ba}, \quad \mathbb{R}_{ab} = \mathbb{R}_{ba}.$$

- *Triangle Inequality*: The effective resistance defines a metric on $\mathbb{G}$ since

$$\boxed{\mathbb{R}_{ac} \le \mathbb{R}_{ab} + \mathbb{R}_{bc}.} \tag{11}$$

Proof. [6] Let v_1 be the voltage function defined as the harmonic extension of $v_1(a) = \mathbb{R}_{ab}$ and $v_1(b) = 0$. Let i_1 denote the corresponding current, a unit flow from a to b.

Let v_2 be the voltage function defined as the harmonic extension of $v_2(b) = 0$ and $v_2(c) = -\mathbb{R}_{ac}$. Let i_2 denote the corresponding current, which in particular is a unit flow from b to c.

Let $v = v_1 + v_2$ and so the corresponding current is $i = i_1 + i_2$. Then i is a unit flow from a to c in the sense of (45), since $i(a) = i_1(a) + i_2(a) = 1 + 0 = 1$, $i(b) = i_1(b) + i_2(b) = -1 + 1 = 0$ and $i(c) = i_1(c) + i_2(c) = 0 - 1 = -1$. Notice that v is harmonic at b since by (6)

$$\Delta v(b) = \Delta v_1(b) + \Delta v_2(b) = -c_b^{-1}(i_1(b) + i_2(b)) = 0.$$

It follows that v is the harmonic extension of $v(a) = \mathbb{R}_{ab} + v_2(a)$ and of $v(c) = v_1(c) - \mathbb{R}_{ac}$. Since i is a *unit* flow from a to c it follows that $v(a) - v(c) = \mathbb{R}_{ac}$.

But $v_2(a) \le 0$ and $v_1(c) \ge 0$ by the maximum principle, and so

$$\mathbb{R}_{ac} = v(a) - v(c) = \mathbb{R}_{ab} + \mathbb{R}_{bc} + v_2(a) - v_1(c) \le \mathbb{R}_{ab} + \mathbb{R}_{bc}.$$ □

We give a random walk proof on page 14 and a proof using trace networks in *** .

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[4] Equality comes from

$$0 = \sum_x \sum_y i_{xy} = \sum_y i_{ay} + \sum_{b\in G}\sum_y i_{by} = i_a + \sum_{b\in G} i_b,$$

where the first equality is because $i_{xy} = -i_{yx}$ and the second because $i_x = 0$ at interior nodes.

[5] If f and g are harmonic so is the linear combination $h = \alpha f + \beta g$ (*superposition principle*). The current corresponding to h along an edge is then the same linear combination of the currents corresponding to f and g along the edge. If f is defined by $f(a) = 1$ and $f(b) = 0$ then the current out of a is $\mathbb{C}_{ab}$. If g is defined by $g(a) = 0$ and $g(b) = 1$ then the current out of b is $\mathbb{C}_{ba}$, and hence out of a is $-\mathbb{C}_{ba}$ by (7). It follows that the current corresponding to $h = f + g$ and out of a is $\mathbb{C}_{ab} - \mathbb{C}_{ba}$. Since $h(x) = 1$ $\forall x$ o the current corresponding to h is zero.

[6] I havn't seen this proof before. It is elementary and gives a precise value for the error term in the triangle inequality.

3.5. **Computing Effective Conductance.**

3.5.1. *Resistances in parallel and in series.* Conductances add in parallel, resistances add in series.

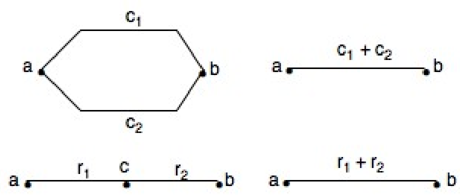

FIGURE 3. Conductances add in parallel, resistances add in series.

In Figure 3 voltages $v(a), v(b)$ are prescribed, as are conductances c_1, c_2 and resistances r_1, r_2. In the top two diagrams the current flowing from a to b is clearly

$$(v(a) - v(b))(c_1 + c_2).$$

In the bottom left diagram direct calculation shows the induced voltage at c is

$$\frac{\frac{1}{r_1}v(a) + \frac{1}{r_2}v(b)}{\frac{1}{r_1} + \frac{1}{r_2}}.$$

Easy calculation then shows the current flowing out of a, and the current flowing into b, is

$$\frac{v(a) - v(b)}{r_1 + r_2}$$

in both of the bottom diagrams.

It follows that in any circuit the two left side diagrams can be replaced by the corresponding right side diagrams when computing induced voltages, current flow or effective resistance. They have the same input-output relations on the nodes $\{a, b\}$ and so we say they are *equivalent* on the nodes $\{a, b\}$. See also Section 11.2. See [DS, p 44] for an example calculation.

3.5.2. *The $\Delta - Y$ transform.* Consider the two networks in Figure 4 with conductances as shown.

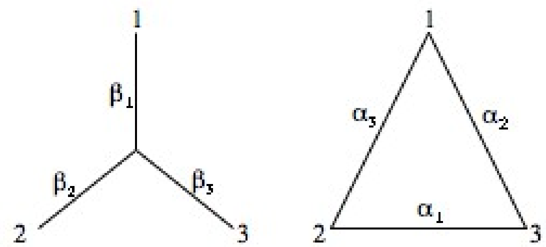

FIGURE 4. The $\Delta - Y$ transform

We say they are *equivalent* if they have the same input output relations on the nodes $\{1, 2, 3\}$.

Proposition 3.1. *The networks in Figure 4 are equivalent on $\{1, 2, 3\}$ iff*

$$\alpha_1 = \frac{\beta_2\beta_3}{\beta_1 + \beta_2 + \beta_3}, \quad \alpha_2 = \frac{\beta_3\beta_1}{\beta_1 + \beta_2 + \beta_3}, \quad \alpha_3 = \frac{\beta_1\beta_2}{\beta_1 + \beta_2 + \beta_3}. \tag{12}$$

Equivalently, iff

$$\beta_i = \frac{S}{\alpha_i}, \quad \textit{where } S = \alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1. \tag{13}$$

Proof. [7] It is sufficient to consider a voltage 1 at node 1 and 0 at nodes 2 and 3, since by symmetry this covers two similar cases by rotating, and then all cases by scaling and summing boundary data.

In the Y diagram, the induced voltage at the central node is $\beta_1/(\beta_1+\beta_2+\beta_3)$. The currents through 1, 2 and 3 then are:

$$i_1 = \beta_1\left(1 - \frac{\beta_1}{\beta_1+\beta_2+\beta_3}\right) = \frac{\beta_1(\beta_2+\beta_3)}{\beta_1+\beta_2+\beta_3}, \quad i_2 = \frac{\beta_1\beta_2}{\beta_1+\beta_2+\beta_3}, \quad i_3 = \frac{\beta_1\beta_3}{\beta_1+\beta_2+\beta_3}$$

In the Δ diagram the corresponding currents are

$$i_1 = \alpha_3 + \alpha_2, \quad i_2 = \alpha_3, \quad i_3 = \alpha_2.$$

Hence

$$\alpha_2 = \frac{\beta_1\beta_3}{\beta_1+\beta_2+\beta_3}, \quad \alpha_3 = \frac{\beta_1\beta_2}{\beta_1+\beta_2+\beta_3},$$

and so

$$\alpha_1 = \frac{\beta_2\beta_3}{\beta_1+\beta_2+\beta_3}$$

by symmetry.

It follows that

$$S := \alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1 = \frac{\beta_1\beta_2\beta_3(\beta_1+\beta_2+\beta_3)}{(\beta_1+\beta_2+\beta_3)^2} = \frac{\beta_1\beta_2\beta_3}{\beta_1+\beta_2+\beta_3},$$

and so $\beta_i = S/\alpha_i$. □

Alternative Proof. From the conductance matrix for Y,

$$A = \begin{bmatrix} -\beta_1 & 0 & 0 & \beta_1 \\ 0 & -\beta_2 & 0 & \beta_2 \\ 0 & 0 & -\beta_3 & \beta_3 \\ \beta_1 & \beta_2 & \beta_3 & -\beta_1-\beta_2-\beta_3 \end{bmatrix}$$

one can use Maple to compute the trace conductance matrix B on Δ via (55). Namely,

$$B = A_{GG} - A_{GH}A_{HH}^{-1}A_{HG} = \begin{bmatrix} -\frac{\beta_1\,(\beta_2+\beta_3)}{\beta_1+\beta_2+\beta_3} & \frac{\beta_1\,\beta_2}{\beta_1+\beta_2+\beta_3} & \frac{\beta_1\,\beta_3}{\beta_1+\beta_2+\beta_3} \\ \frac{\beta_1\,\beta_2}{\beta_1+\beta_2+\beta_3} & -\frac{\beta_2\,(\beta_1+\beta_3)}{\beta_1+\beta_2+\beta_3} & \frac{\beta_2\,\beta_3}{\beta_1+\beta_2+\beta_3} \\ \frac{\beta_1\,\beta_3}{\beta_1+\beta_2+\beta_3} & \frac{\beta_2\,\beta_3}{\beta_1+\beta_2+\beta_3} & -\frac{\beta_3\,(\beta_1+\beta_2)}{\beta_1+\beta_2+\beta_3} \end{bmatrix}.$$

□

4. Random Walk Interpretation of a Graph

4.1. **Definition of the RW on $\mathbb{G}$.** This is characterised by defining the probability of moving from x to y to be

$$p_{xy} = \frac{c_{xy}}{c_x}. \tag{14}$$

Thus *rows* of the *transition matrix* $P := (p_{xy})_{x\in V, y\in V}$ are probability vectors, with row x containing the probabilities corresponding to leaving node x.

The RW always begins at time 0*, and its position at time* $n \geq 0$ *is denoted by* X_n*.*

[7] This is less tedious than computing the effective resistances between the pairs 12, 23 and 31.

4.2. **Markov Chain.** The random walk is a markov chain with state space V and transition probabilities p_{xy}.

- The markov chain is *irreducible* since $\mathbb{G}$ is connected, and so any state can be reached from any other state.
- Hence by standard markov chain theory there is a *unique steady state distribution* $(w_x)_{x\in V}$.
- As for any markov chain, if $(u_x)_{x\in V}$ is the probability distribution at one time then $(v_x)_{x\in V}$, where $v_x = \sum_y u_y p_{yx}$, gives the probability distribution at the next time.
- It follows by direct calculation that the *steady state distribution w_x is proportional to c_x*,[8] and so

$$w_x = \frac{c_x}{C}. \tag{15}$$

- The markov chain is *reversible*, i.e.

$$w_x p_{xy} = w_y p_{yx} \quad \forall x, y \in V.$$

- *Interpretation of reversibility*: One cannot tell the direction of time by watching the RW evolve from the steady state distribution w. More precisely

$$\begin{aligned} P(xy) &:= P(X_0 = x)\, P(X_1 = y | X_0 = x) = w_x p_{xy} \\ &= w_y p_{yx} = P(X_0 = y)\, P(X_1 = x | X_0 = y) =: P(yx). \end{aligned}$$

  Similarly

$$\begin{aligned} P(x_0 x_1 \dots x_n) &= w_{x_0} p_{x_0 x_1} p_{x_1 x_2} \cdots p_{x_{n-1} x_n} \\ &= w_{x_n} p_{x_n x_{n-1}} \cdots p_{x_2 x_1} p_{x_1 x_0} = P(x_n \dots x_1 x_0). \end{aligned}$$

  The following equivalent version does not explicitly refer to the steady state distribution. For any path $x_0 x_1 \dots x_n$ and beginning at each end respectively,

$$c_{x_0} P^{x_0}(x_0 x_1 \dots x_n) = c_{x_n} P^{x_n}(x_n \dots x_1 x_0).$$

- Conversely, any reversible markov chain is the RW given by a graph, provided we allow $c_{xx} > 0$.[9]
- A natural measure on V is $(c_x)_{x\in V}$. The corresponding *transition density* is defined for $x \sim y$ by

$$p(x,y) := \frac{P^x(X_1 = y)}{c_y} = \frac{c_{xy}}{c_x c_y} = \frac{P^y(X_1 = x)}{c_x} = p(y,x).$$

  Reversibility thus is equivalent to symmetry of the transition density $p(x,y)$ (but *not* of p_{xy}).

4.3. **Two Functions Associated with the RW.** Suppose $\emptyset \neq G \subset V$, $a \in V \setminus G$ and $x \in V$.

We introduce two important functions h and g which equal 0 on G and are harmonic at $x \notin \{a\} \cup G$. By the scaling comments in Section 3.3, they are multiples of each other, of the voltage v in (1) and (2), and of the Green's function discussed later. We note the consequences in Section 5.

First some notation.

Notation: By T_a (T_G) is meant *the time of first arrival at a (G)*. This is from some designated starting point, which in (16) is x.

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[8]Since if $w_x := c_x/C$ then

$$\sum_y w_y p_{yx} = \sum_y \frac{c_y}{C}\frac{c_{yx}}{c_y} = \sum_y \frac{c_{yx}}{C} = \sum_y \frac{c_{xy}}{C} = \frac{c_x}{C} = w_x.$$

Hence w_x is the steady state distribution by uniqueness.

[9]Suppose the markov chain is p and the steady state distribution is w. Then $w_x p_{xy} = w_y p_{yx}$ $\forall x, y$. Define $c_{xy} = w_x p_{xy}$ and note $c_{xy} = c_{yx}$. Moreover,

$$\frac{c_{xy}}{c_x} = \frac{c_{xy}}{\sum_z c_{xz}} = \frac{w_x p_{xy}}{\sum_z w_x p_{xz}} = p_{xy},$$

and so this gives the original markov chain.

By $V_{x\to G}$ is meant *the number of visits to x before arriving at G*. This is from some designated starting point, which in (17) is a. If x is the starting point then *the initial visit at time 0 is included* as a visit to x. If $x \in G$ then $V_{x\to G} = 0$.

Let

$$\boxed{h(x) = P^x(T_a < T_G),} \tag{16}$$

$$\boxed{g(x) = \frac{1}{c_x}\,\mathbb{E}^a(V_{x\to G}).} \tag{17}$$

Proposition 4.1.

$$\begin{gathered} h(a) = 1, \quad h(b) = 0 \textit{ if } b \in G, \\ g(a) = \frac{1}{c_a}\mathbb{E}^a V_{a\to G}, \quad g(b) = 0 \textit{ if } b \in G. \end{gathered} \tag{18}$$

Moreover,

$$\Delta\, h(x) = 0, \quad \Delta\, g(x) = 0, \quad x \notin \{a\} \cup G. \tag{19}$$

Proof. The boundary conditions are immediate.

Because the start from x is *followed* by a visit to some $y \sim x$,

$$h(x) = \sum_y p_{xy} P^y(T_a < T_G) = \sum_y \frac{c_{xy}}{c_x}\, h(y); \tag{20}$$

and because every visit to x is *preceded* by a visit to some $y \sim x$,[10]

$$\begin{aligned} g(x) &:= \frac{1}{c_x}\mathbb{E}^a V_{x\to G} = \frac{1}{c_x}\sum_y p_{yx}\,\mathbb{E}^a V_{y\to G} \\ &= \sum_y \frac{1}{c_x}\frac{c_{yx}}{c_y}\, c_y\, g(y) = \sum_y \frac{c_{xy}}{c_y} g(y). \end{aligned} \tag{21}$$

□

Note that *reversibility is required for the proof that g is harmonic*, but not so for h.

For later purposes, keeping in mind the scaling comments in Section 3.3, we next compute the Laplacians at a.

Let T_a^+ denotes the first *return* time to a, i.e. the time of the first visit to a *after* time 0 or equivalently the first visit in positive time. Then

$$P^a(T_G < T_a^+) \tag{22}$$

denotes the probability the walk reaches G from a without returning to a. It is also called the *escape probability* from a to G.

Proposition 4.2.

$$-\,\Delta\, h(a) = P^a(T_G < T_a^+), \quad -\,\Delta\, g(a) = \frac{1}{c_a}. \tag{23}$$

Proof. For h,

$$\begin{aligned} -\,\Delta\, h(a) &= \sum_x \big(h(a) - h(x)\big)p_{ax} \\ &= \sum_x \big(1 - P^x(T_a < T_G)\big)p_{ax} \\ &= \sum_x p_{ax}P^x(T_G < T_a) = P^a(T_G < T_a^+). \end{aligned}$$

[10]More precisely, fix $x \notin \{a\} \cup G$. Then

$$\begin{aligned} g(x) &= \frac{1}{c_x}\mathbb{E}^a(V_{x\to G}) = \frac{1}{c_x}\sum_{n\geq 1} P^a(X_n = x) \quad (\text{as } x \neq a) \\ &= \frac{1}{c_x}\sum_{n\geq 1}\sum_{y\sim x} P^a(X_{n-1} = y)p_{yx} = \frac{1}{c_x}\sum_{y\sim x}\frac{c_{yx}}{c_y}\,\mathbb{E}^a(V_{y\to G}) = \sum_{y\sim x}\frac{c_{xy}}{c_x} g(y). \end{aligned}$$

For g use the same calculation as in (21) after noting

$$g(a) := \frac{1}{c_a}\mathbb{E}^a V_{a\to G} = \frac{1}{c_a} + \frac{1}{c_a}\sum_y p_{ya}\,\mathbb{E}^a V_{y\to G},$$

where the second equality is because every visit to a except for the first is preceded by a visit to some $y \sim a$. □

We give a random walk proof of the following. It is also a consequence of the electric network interpretation; see the sentence following (30).

Proposition 4.3. *For $a \notin G$,*

$$\boxed{\mathbb{E}^a V_{a\to G} = 1/P^a(T_G < T_a^+).} \tag{24}$$

Proof. Let

$$\begin{aligned} p &= P^a(T_G < T_a^+) \quad \text{(i.e. “success”)},\\ 1-p &= P^a(T_a^+ < T_G) \quad \text{(i.e. “failure”)}. \end{aligned}$$

The number of “failures” before a success is a geometric distribution with parameter p. That is, the probability of exactly n returns to a before reaching G, or equivalently exactly $n+1$ visits, is $(1-p)^n p$. Hence the expected number of visits to a before reaching G is $1+$ the mean of the geometric distribution, and so by a standard result[11] is $1 + (1-p)/p = 1/p$. □

5. Analysis, Connections with EN and RW

Suppose $\emptyset \neq G \subset V$, $a \in V \setminus G$ and $x \in V$.

5.1. Green's Function.

5.1.1. *Definition.* Motivated by the classical situation,[12] *Green's function centred at a* (for the set G) is the unique function $\mathcal{G}_a^G = \mathcal{G}_a : V \to \mathbb{R}$ given by

$$\begin{aligned} &\boxed{-\Delta\,\mathcal{G}_a(x) = \frac{1}{c_a}\mathcal{X}_a(x),} \quad x \in V \setminus G,\\ &\mathcal{G}_a(x) = 0 \quad x \in G, \end{aligned} \tag{25}$$

where $\mathcal{X}_a$ is the characteristic function equal to 1 at a and 0 elsewhere. Uniqueness, and hence existence, follow as in Section 3.2. Note that $\dfrac{1}{c_a}\mathcal{X}_a$ is the natural choice as the “Dirac measure” centred at a since it is supported on a and integrates to 1 w.r.t. the natural measure $(c_x)_{x\in V}$.

[11] To see this directly let Z denote the number of visits to a before first reaching G. Then

$$\begin{aligned} \mathbb{E}(Z) &= \sum_{n\ge 0}(n+1)(1-p)^n p = -p\sum_{n\ge 0}\frac{d}{dp}(1-p)^{n+1}\\ &= -p\frac{d}{dp}\sum_{n\ge 0}(1-p)^{n+1} = -p\frac{d}{dp}\left(\frac{1-p}{p}\right) = \frac{1}{p}. \end{aligned}$$

[12] Suppose $a \in \Omega$. Then the Green function $\mathcal{G}_a^{\partial\Omega}(x) = \mathcal{G}_a(x)$ is characterised for $x \in \overline{\Omega}$ by

$$\begin{aligned} -\Delta\,\mathcal{G}_a(x) &= \delta_a(x) \quad \text{if } x \in \Omega, \quad \text{(Dirac delta measure, equality in distributional sense)}\\ \mathcal{G}_a(x) &= 0 \quad \text{if } x \in \partial\Omega. \end{aligned}$$

In particular,

$$\mathcal{G}_a(x) = \begin{cases} -\frac{\ln|x-a|}{2\pi} + \psi(x) & n = 2,\\ \frac{c_n}{|x-a|^{n-2}} + \psi(x) & n \ge 3,\ (c_3 = \frac{1}{4\pi}) \end{cases}$$

where $\psi(x)$ is the unique harmonic function on Ω such that $\mathcal{G}_a(x) = 0$ if $x \in \partial\Omega$.

5.1.2. *Symmetry.* This is analogous to the proof in the classical setting.[13]

$$\boxed{\mathcal{G}_b(a) = \mathcal{G}_a(b).} \quad a, b \notin G \tag{26}$$

Proof. From Green's identity (41), which is just an algebraic calculation,

$$(\Delta\,\mathcal{G}_a, \mathcal{G}_b) = (\mathcal{G}_a, \Delta\,\mathcal{G}_b),$$

where each side is the natural inner product with the weighting c_x at x. The left side from (25) is $c_a \mathcal{G}_b(a)\dfrac{1}{c_a} = \mathcal{G}_b(a)$ and the right side is similarly $\mathcal{G}_a(b)$. □

5.1.3. *Poisson's equation.* Analogously to the PDE setting,[14] the solution of

$$\begin{aligned} -\Delta\, u(x) &= f(x) \quad x \in V \setminus G, \\ u(x) &= 0 \quad x \in G, \end{aligned} \tag{27}$$

is given by

$$\boxed{u(x) = \sum_{y \in V \setminus G} \mathcal{G}_y(x) f(y)\, c_y.} \tag{28}$$

Proof. If $x \in G$ then

$$\sum_{y \in V \setminus G} \mathcal{G}_y(x) f(y)\, c_y = 0,$$

and if $x \in V \setminus G$ then

$$-\Delta_x \left(\sum_{y \in V \setminus G} \mathcal{G}_y(x) f(y)\, c_y \right) = \sum_{y \in V^\circ} \frac{\mathcal{X}_y(x)}{c_y} f(y) c_y = f(x).$$ □

5.2. **Summary of Connections so far.** We defined the following functions on V, harmonic for $x \notin G$:

$$\begin{aligned} v(x) &= \text{voltage given by (1), (2)}, \\ h(x) &= P^x(T_a < T_G), \\ g(x) &= \frac{1}{c_x} \mathbb{E}^a V_{x \to G}, \\ \mathcal{G}_a^G(x) &= \text{Green function centred at } a \text{ given by (25)}. \end{aligned}$$

[13] Symmetry of Green's function is shown for $a \neq b \in \Omega$ as follows:

$$\mathcal{G}_a(b) - \mathcal{G}_b(a) = \int_\Omega \mathcal{G}_b\, \Delta\, \mathcal{G}_a - \mathcal{G}_a\, \Delta\, \mathcal{G}_b = \int_{\partial\Omega} \mathcal{G}_b \frac{\partial \mathcal{G}_a}{\partial \nu} - \mathcal{G}_a \frac{\partial \mathcal{G}_b}{\partial \nu} = 0$$

The first and third equalities are from the definition of $\mathcal{G}_a$ and $\mathcal{G}_b$. The second is by Green's formula. Strictly, one should integrate over $\Omega \setminus (B_\epsilon(a) \cup B_\epsilon(a))$ and let $\epsilon \to 0$.

[14] The solution of

$$\begin{aligned} -\Delta\, u &= f \quad \text{in } \Omega \\ u &= 0 \quad \text{on } \partial\Omega \end{aligned}$$

is given by the following "weighted sum" of Green functions:

$$u(x) = \int_\Omega \mathcal{G}_y(x) f(y)\, dy.$$

In fact formally, and made precise by a careful limiting argument, one has

$$\int_\Omega \mathcal{G}_y(x) f(y)\, dy = \int_\Omega \Delta_x\, \mathcal{G}_y(x) f(y)\, dy = \int_\Omega \delta_x(y) \mathcal{G}_y(x) f(y)\, dy = f(x).$$

Moreover,

$$\begin{aligned} v(a) &= 1 & -c_a\,\Delta\, v(a) &= i_a = \mathbb{C}_{aG} \\ h(a) &= 1 & -c_a\,\Delta\, h(a) &= c_a P^a(T_G < T_a^+) \\ g(a) &= \frac{1}{c_a}\mathbb{E}^a V_{a\to G} & -c_a\,\Delta\, g(a) &= 1 \\ \mathcal{G}_a^G(a) &= (\text{no nice analytic expr}) & -c_a\,\Delta\, \mathcal{G}_a^G(a) &= 1 \end{aligned}$$

In particular,

$$\boxed{v(x) = P^x(T_a < T_G) = \mathbb{C}_{aG}\,\frac{\mathbb{E}^a V_{x\to G}}{c_x} = \mathbb{C}_{aG}\,\mathcal{G}_a^G(x).} \tag{29}$$

The other information we extract, by using the values of these functions and their Laplacians at a together with the scaling comments in Section 3.3, is

$$\boxed{\mathbb{R}_{aG} = \frac{1}{i_a} = \frac{1}{c_a P^a(T_G < T_a^+)} = \frac{\mathbb{E}^a V_{a\to G}}{c_a} = \mathcal{G}_a^G(a).} \tag{30}$$

Note that the third equality above follows directly from the fact that g and h are multiples of each other, and is independent of the direct proof in Proposition 4.3.

Consider a non empty set of "boundary" nodes $B = \partial V \subset V$. This generalises the situation in Sections 3, 4 and 5, where $\partial V = \{a, b\}$ or $\partial V = \{a\}$.

5.3. **Currents and Traverses along Edges.** There is also a connection between the current along an edge and the net number of traverses along the edge.

For this let $Tr_{xy\to G}$ denote *the net number of traverses from x to y before reaching G*.

Proposition 5.1. *Suppose a voltage is applied at a and zero voltage is applied at G, so that* unit *current flows into V at a. Then if i_{xy} denotes the (signed) current along the edge xy,*

$$\boxed{i_{xy} = \mathbb{E}^a Tr_{xy\to G}.} \tag{31}$$

Proof. The voltage $v(x)$ for unit current flow is $g(x)$ as defined in (17), since both $v(x)$ and $g(x)$ are characterised by $-c_a\,\Delta\, v(a) = -c_a\,\Delta\, g(a) = 1$, see (6) and (23). Hence

$$\begin{aligned} i_{xy} &= (g(x) - g(y))\, c_{xy} \\ &= \frac{c_{xy}}{c_x}\mathbb{E}^a(V_{x\to G}) - \frac{c_{xy}}{c_y}\mathbb{E}^a(V_{y\to G}) = p_{xy}\mathbb{E}^a(V_{x\to G}) - p_{yx}\mathbb{E}^a(V_{y\to G}) \\ &= \mathbb{E}^a(\text{no. of traverses from } x \text{ to } y \text{ before } G) - \mathbb{E}^a(\text{no. of traverses from } y \text{ to } x \text{ before } G) \\ &= \mathbb{E}^a(\textit{net} \text{ no. of traverses from } x \text{ to } y \text{ before } G) = \mathbb{E}^a Tr_{xy\to G}. \end{aligned}$$

□

Note that for $x \sim y$,

$$g(x) := \frac{1}{c_x}\mathbb{E}^a V_{x\to G} = \frac{1}{c_{xy}}\mathbb{E}^a(\text{no. of traverses from } x \text{ to } y \text{ before } G). \tag{32}$$

The right side is the normalised no. of traverses in the direction xy, *not* the net no. of traverses.

5.4. **Dirichlet Problem.** Suppose $G \subset V$. (There is here no role played by a.)

Analogous to the classical P.D.E. problem[15] suppose $h : G \to \mathbb{R}$ is prescribed. Then there exists a unique extension of h to $V \setminus G$ satisfying

$$\Delta\, h(x) = 0 \quad x \notin G. \tag{33}$$

Uniqueness, and hence existence, follow as in Section 3.2.

RW Interpretation: (This is analogous to the case of Brownian Motion in the P.D.E. setting.[16]) The solution h is the expected "payoff":

$$\boxed{h(x) = \mathbb{E}^x h(X_{T_G}).} \tag{34}$$

---

[15]Let $\Omega \subset \mathbb{R}^n$ be non empty bounded and open with a "nice" boundary $\partial\Omega$, e.g. C^1. Suppose $h : \partial V \to \mathbb{R}$ is prescribed and also "nice", e.g. C^1. Then there is a unique harmonic extension of h to Ω.

[16]If X is Brownian motion on Ω and $T_{\partial\Omega}$ is the time of first reaching the boundary, then

$$\phi(x) = \mathbb{E}^x\phi(X_{T_{\partial\Omega}}).$$

where T_G is the time of first arrival at G. The fact that h agrees with the given boundary values, and is harmonic on $V \setminus G$, is given by the same argument as for Proposition 4.1.

EN Interpretation: It also follows as in Sections 3.1 and 3.2 that we can regard h as giving the voltage at each node in V for the prescribed voltages on G.

6. Crossing Time, Einstein Relation

6.1. Crossing Time Theorem. Note that we do not normally have $\mathbb{E}^a T_b = \mathbb{E}^b T_a$. For example, consider a triangular network with nodes $V = \{a, b, c\}$. Suppose $c_{ab} = c_{ac} = \epsilon$ and $c_{bc} = 1$. Suppose ϵ is small.

If the walk starts from a then 1/2 of the time it will go directly to b. The other 1/2 it will go to c and then with probability $1 - \epsilon$ go directly to c. So

$$\mathbb{E}^a T_b = \frac{1}{2} + \frac{1}{2} 2 + O(\epsilon) = \frac{3}{2} + O(\epsilon).$$

If the walk starts from b then it will bounce back and forth many times between b and c before reaching a. In fact for any fixed n, $\mathbb{E}^b T_a = n - O(\epsilon) >> \mathbb{E}^a T_b$.

The following was first proved in [Chandra et al, 1989].

$$\boxed{\mathbb{E}^a T_b + \mathbb{E}^b T_a = \mathbb{R}_{ab}\, C(\mathbb{G}).} \tag{35}$$

Proof. [17] One has

$$\mathbb{E}^a T_b + \mathbb{E}^b T_a = \sum_x \left(\mathbb{E}^a V_{x\to b} + \mathbb{E}^b V_{x\to a}\right)$$

From (30) and the previously proved $\mathbb{R}_{ab} = \mathbb{R}_{ba}$, $c_x^{-1}\left(\mathbb{E}^a V_{x\to b} + \mathbb{E}^b V_{x\to a}\right)$ is harmonic with boundary values $\mathbb{R}_{ab}$ at a and b. Hence

$$c_x^{-1}\left(\mathbb{E}^a V_{x\to b} + \mathbb{E}^b V_{x\to a}\right) = \mathbb{R}_{ab}$$

for all $x \in V$. It follows that

$$\mathbb{E}^a T_b + \mathbb{E}^b T_a = \mathbb{R}_{ab}\, C(\mathbb{G}).$$

□

6.2. Einstein Relation. The left side of (35) is the expected *commute time* from a to b and back, $\mathbb{R}_{ab}$ is the distance between a and b in the resistance metric, and $C = C(\mathbb{G})$ is the total mass of $\mathbb{G}$ (see page 3). This gives the multiplicative version of the *Einstein relation*:

$$\boxed{\text{Time} = \text{Distance} \times \text{Mass}.} \tag{36}$$

We often have a "scale" or parameter x, such as Euclidean distance, where for appropriate exponents

$$\text{Time} \sim x^{d_w},\ \text{Distance} \sim x^{d_r},\ \text{Mass} \sim x^{d_f}. \tag{37}$$

More precisely: "Time" is the commute time between points which are separated by Euclidean distance $\approx x$, "Distance" is the resistance metric distance between points which are separated by Euclidean distance $\approx x$, and "Mass" is the mass of a ball whose Euclidean diameter is $\approx x$.

The exponents are called the *walk dimension* d_w, the *resistance dimension* d_r, and the *fractal dimension* d_f.

From (36) and (37) one gets the usual form of the *Einstein relation*:

$$\boxed{d_w = d_r + d_f.} \tag{38}$$

[17] I have not seen quite this proof before, but it is natural within the current development. It only uses the facts that $c_x^{-1}\mathbb{E}^a V_{x\to b}$ is harmonic except at a and b, its value at a is $\mathbb{R}_{ab}$ and $\mathbb{R}_{ab}$ is symmetric.

6.3. **Triangle Inequality.** Here is a random walk proof of the triangle inequality for the resistance metric. This also gives the error term, see Footnote 18.

Proof. Clearly[18]

$$\mathbb{E}^a T_c \leq \mathbb{E}^a T_b + \mathbb{E}^b T_c.$$

Switching a and c and adding,

$$\mathbb{E}^a T_c + \mathbb{E}^c T_a \leq \mathbb{E}^a T_b + \mathbb{E}^b T_a + \mathbb{E}^b T_c + \mathbb{E}^c T_b.$$

Dividing both sides by $C(\mathbb{G})$ and using the crossing time inequality (35) gives

$$\mathbb{R}_{ac} \leq \mathbb{R}_{ab} + \mathbb{R}_{bc}.$$ □

7. Dirichlet Energy

7.1. **Definitions.**

- For $f, g : V \to \mathbb{R}$ define the *Dirichlet Energy* of f, and corresponding bilinear form, by

$$\boxed{\mathcal{E}(f) = \frac{1}{2}\sum_{x,y} c_{xy}\big(f(x) - f(y)\big)^2,}$$

$$\boxed{\mathcal{E}(f,g) = \frac{1}{2}\sum_{x,y} c_{xy}\big(f(x) - f(y)\big)\big(g(x) - g(y)\big).} \tag{39}$$

  The $\frac{1}{2}$ arises since the energy is "sitting" on edges, but each edge appears twice in the sum.
- *EN Interpretation*: If f is a voltage then $i_{xy} = c_{xy}(f(x) - f(y))$ is the current along the directed edge xy and the first summand in (39),

$$c_{xy}\big(f(x) - f(y)\big)^2 = i_{xy}\big(f(x) - f(y)\big) = i_{xy}^2 r_{xy}, \tag{40}$$

  is the "energy dissipation" on the edge xy. The Dirichlet energy $\mathcal{E}(f)$ is the "energy dissipation" of the network, or equivalently the energy required to keep the network flowing.

7.2. **Green's Identity.** (This is analogous to the P.D.E. case.)

$$\boxed{\mathcal{E}(f,g) = -(\Delta\, f, g) = -(f, \Delta\, g),} \tag{41}$$

where $(h,k) = \sum_x h(x)k(x)c_x$ denotes the L^2 inner product with the natural measure $(c_x)_{x\in V}$.

In particular, Δ *is self-adjoint.*

Note that (41) expresses the energy $\mathcal{E}(f)$ as a sum of contributions at *nodes* rather than edges, and the contribution is 0 at any node where $\Delta\, f = 0$ or $f = 0$.

Moreover, suppose f and g are the harmonic extensions of applied voltages on G. Let i^f and i^g denotes the corresponding currents flowing into V on G. Then (41) can be written

$$\boxed{\mathcal{E}(f,g) = \sum_{x\in G} f(x)\, i^g(x) = \sum_{x\in G} i^f(x)\, g(x).} \tag{42}$$

---

[18]Using an obvious notation, $T_c^a = T_b^a + T_c^b$ if $T_b^a \leq T_c^a$ and $T_c^a = T_b^a - T_b^c < T_b^a + T_c^b$ if $T_c^a < T_b^a$. So $T_c^a \leq T_b^a + T_c^b$ in both cases.

More precisely,

$$\begin{aligned}\mathbb{E}^a T_c &= P^a(T_b \leq T_c)\,\mathbb{E}^a(T_c \mid T_b \leq T_c) + P^a(T_c < T_b)\,\mathbb{E}^a(T_c \mid T_c < T_b)\\ &= P^a(T_b \leq T_c)\,(\mathbb{E}^a T_b + \mathbb{E}^b T_c) + P^a(T_c < T_b)\,\big(\mathbb{E}^a T_b + \mathbb{E}^b T_c - (\mathbb{E}^b T_c + \mathbb{E}^c T_b)\big)\\ &= (\mathbb{E}^a T_b + \mathbb{E}^b T_c) - P^a(T_c < T_b)\mathbb{E}_c^b,\end{aligned}$$

where $C_c^b = C_b^c := T_c^b + T_b^c$ is the *commute time* from to c and back to b, and $\mathbb{E}_c^b := \mathbb{E}C_c^b$.

Proof of (41).

$$\begin{aligned}\mathcal{E}(f,g) &= \frac{1}{2}\sum_{x,y} c_{xy} f(x)\big(g(x)-g(y)\big) - \frac{1}{2}\sum_{x,y} c_{xy} f(y)\big(g(x)-g(y)\big)\\ &= \frac{1}{2}\sum_{x,y} c_{xy} f(x)\big(g(x)-g(y)\big) + \frac{1}{2}\sum_{x,y} c_{yx} f(x)\big(g(x)-g(y)\big)\\ &= \sum_{x,y} c_{xy} f(x)\big(g(x)-g(y)\big) = -\sum_{x} f(x)\,\Delta\, g(x)\, c_x.\end{aligned}$$

□

7.3. **Thompson's First Principle.** (This is essentially just a particular case of the standard Dirichlet principle.)

Proposition 7.1. *Suppose $h|_{\partial V}$ is prescribed and $\Delta\, h = 0$ on V°. Then*

$$\boxed{\mathcal{E}(h) = \min\{\mathcal{E}(f) : f = h \text{ on } \partial V\}.} \tag{43}$$

In the case $\partial V = \{a,b\}$, $\Delta\, h = 0$ on V°, $h(a) = 1$ and $h(b) = 0$, then

$$\boxed{\mathcal{E}(h) = \min\{\mathcal{E}(f) : f(a) = 1, f(b) = 0\} = \mathbb{C}_{ab}.} \tag{44}$$

The following proof that the Dirichlet minimiser is harmonic is analogous to the P.D.E. situation.[19] The proof that the Dirichlet energy of the harmonic function is $\mathbb{C}_{ab}$ follows essentially directly from Green's Identity and the definition of $\mathbb{C}_{ab}$.

Proof. Suppose $\phi = 0$ on ∂V. Then

$$\begin{aligned}\mathcal{E}(h+\phi, h+\phi) &= \mathcal{E}(h,h) + 2\mathcal{E}(h,\phi) + \mathcal{E}(\phi,\phi) \quad \text{(bilinearity)}\\ &= \mathcal{E}(h,h) - 2(\Delta\, h,\phi) + \mathcal{E}(\phi,\phi) \quad \text{(Green's identity)}\\ &= \mathcal{E}(h,h) + \mathcal{E}(\phi,\phi) \quad (\text{since } \Delta\, h = 0 \text{ on } V^\circ \text{ and } \phi = 0 \text{ on } \partial V)\\ &\geq \mathcal{E}(h,h)\end{aligned}$$

Equality holds in the last line iff ϕ is constant, i.e. $\phi \equiv 0$.

This shows that h is the unique minimiser.

In the second case, by Green's identity (41), $\mathcal{E}(h) = -c_a\,\Delta\, h(a)$. But $-c_a\,\Delta\, h(a)$ is the current i_a under the voltage h, by (6). Since $h(a) = 1$ and $h(b) = 0$, it follows $i_a = \mathbb{C}_{ab}$ by the definition of C_{ab} in (10). □

7.4. **Flows.** The following definitions are motivated by the properties of a current induced by a voltage applied on ∂V, over a network with possibly unknown conductances.

Fix $\partial V \subset V$. A *flow* is a function j on the directed edges of $\mathbb{G}$ such that

$$\begin{aligned} j_{xy} &= 0 \text{ for } x \not\sim y\\ j_{xy} &= -j_{yx}\\ \sum_y j_{xy} &= 0 \text{ if } x \in V^\circ.\end{aligned} \tag{45}$$

---

[19](See footnote 15.) The harmonic extension to Ω of $h : \partial\Omega \to \mathbb{R}$ is the unique minimiser in $H^1(\Omega)$ of $\mathcal{E}(f) := \frac{1}{2}\int_\Omega |Df^2$ with the prescribed boundary conditions. Existence comes from taking a minimising sequence and standard P.D.E. arguments to show the sequence converges in the H^1 sense.

For here, just note that if h is harmonic and $\phi \in H^1_0(\Omega)$ then

$$\begin{aligned}\mathcal{E}(h+\phi) &= \frac{1}{2}\int_\Omega |Dh|^2 + \int_\Omega Dh\cdot D\phi + \frac{1}{2}\int_\Omega |D\phi|^2\\ &= \mathcal{E}(h) - \int_\Omega \Delta\, h\cdot\phi + \mathcal{E}(\phi)\\ &= \mathcal{E}(h) + \mathcal{E}(\phi)\\ &\geq \mathcal{E}(h), \text{ with equality iff } \phi = 0 \text{ a.e.}\end{aligned}$$

So h is indeed the unique minimiser.

The *flow out of* x for $x \in V$ (or equivalently *into* $\mathbb{G}$ at x) is

$$j_x := \sum_y j_{xy}.$$

Hence $j_x = 0$ if $x \in V^\circ$.

The *energy* of j is

$$\boxed{\mathcal{E}(j) := \frac{1}{2}\sum_{x,y} j_{xy}^2 c_{xy}^{-1}.} \tag{46}$$

This is the same as the energy of v if j is the current corresponding to the voltage v.

If $\partial V = \{a, b\}$ and $j_a = -j_b = 1$, then j is a *unit flow from a to b*.

7.5. **Properties of Flows.** Suppose j is a flow and $w : V \to \mathbb{R}$. Then for any function w_x ,

$$\sum_{x\in\partial V} j_x = 0, \quad j_a = -j_b \text{ if } \partial V = \{a, b\},$$

$$\boxed{\frac{1}{2}\sum_{x,y}(w_x - w_y)j_{xy} = \sum_{x\in\partial V} w_x j_x.} \tag{47}$$

Note the latter is just Green's identity (41) if $w = f$ and j is the current corresponding to the voltage g.

Proof. The first line follows from the second with $w_x \equiv 1$.

For the second,

$$\begin{aligned}\frac{1}{2}\sum_{x,y}(w_x - w_y)j_{xy} &= \frac{1}{2}\sum_{x,y} w_x j_{xy} - \frac{1}{2}\sum_{x,y} w_y j_{xy} = \frac{1}{2}\sum_{x,y} w_x j_{xy} - \frac{1}{2}\sum_{x,y} w_x j_{yx}\\ &= \sum_{x,y} w_x j_{xy} = \sum_x w_x j_x\\ &= \sum_{x\in\partial V} w_x j_x.\end{aligned}$$

□

7.6. **Thompson's Second Principle.**

Proposition 7.2. *Suppose i is the unit flow corresponding to a voltage v applied to a and b, with $v(b) = 0$. Then*

$$\boxed{\mathcal{E}(i) = \min\{\mathcal{E}(j) : j \textit{ is a unit flow from } a \textit{ to } b\} = \mathbb{R}_{ab}.} \tag{48}$$

The proof of the following is analogous to that for Thompson's first principle. The value of the minimiser follows from that in the first principle by a simple scaling argument.

Proof. For feasible j let $j_{xy} = i_{xy} + d_{xy}$. Then d_{xy} is a flow with $d_a = d_b = 0$. Hence

$$\begin{aligned}\mathcal{E}(j) &= \frac{1}{2}\sum_{x,y}(i_{xy} + d_{xy})^2 r_{xy}\\ &= \frac{1}{2}\sum_{x,y} i_{xy}^2 r_{xy} + \sum_{x,y} i_{xy}d_{xy}r_{xy} + \frac{1}{2}\sum_{x,y} d_{xy}^2 r_{xy}.\end{aligned}$$

From (47) the second term on the right equals

$$\sum_{x,y}(v(x) - v(y))d_{xy} = 2v(a)d_a + 2v(b)d_b = 0.$$

Hence

$$\mathcal{E}(j) \geq \mathcal{E}(i),$$

with equality iff $d = 0$, i.e. $i = j$.

If a unit current flows from a to b the applied voltage is $\mathbb{R}_{ab}$, by the definition of $\mathbb{R}_{ab}$. But if the applied voltage is 1 then the Dirichlet energy is $\mathbb{C}_{ab}$ from Thompson's first principle. Hence if the applied voltage is $\mathbb{R}_{ab}$ the Dirichlet energy is multiplied by $\mathbb{R}_{ab}^2$ from the definition (39) of Dirichlet energy, since all voltages will then also be scaled up by $\mathbb{R}_{ab}$. That is,

$$\mathcal{E}(i) = \mathbb{R}_{ab}^2 \mathbb{C}_{ab} = \mathbb{R}_{ab}.$$

□

7.7. **Alternative Expression for Dirichlet Energy.** It is convenient for calculations and otherwise to write $\mathcal{E}(f,g)$ as a bilinear form in the standard manner, using a positive semi definite matrix. This will give a slightly different, but more useful, way of expressing the formula for harmonic extensions and for Green's function. In particular, we do this in Section 11 concerning trace networks.

Note:

(1) $\mathcal{E}(f,g) = \frac{1}{2}\sum_{x,y} c_{xy}(f_x - f_y)(g_x - g_y)$ defines a positive semi definite symmetric bilinear form, and $\mathcal{E}(f,f) = 0$ iff f constant.[20]
(2) The terms c_{xx} are zero. Replacing c_{xx} by any other value will not alter the value of $\mathcal{E}(f,g)$.
(3) As for any (positive semi definite) symmetric bilinear form, $\mathcal{E}(f,g) = f^T B g = g^T B f$ for some (positive semi definite) symmetric matrix B.
(4) Formula (50) expresses the energy as a sum of contributions from *nodes* rather than edges.

In the following note that $-A$ is the positive semi definite matrix. A is often called a *generator matrix*.[21]

Proposition 7.3. *Let $A = (a_{xy})$ for $x, y \in V$, where*

$$a_{xy} = \begin{cases} c_{xy} & x \neq y \\ -c_x & x = y \end{cases}. \tag{49}$$

Then

$$\boxed{\mathcal{E}(f,g) = -\sum_{x,y} a_{xy} f_x g_y} \quad (= -f^T A g = -g^T A f). \tag{50}$$

Proof.

$$\begin{aligned}
\mathcal{E}(f,g) &= \sum_{x,y} c_{xy} f_x g_x - \sum_{x,y} c_{xy} f_x g_y \quad (\text{since } c_{xy} = c_{yx}) \\
&= \sum_{x} c_x f_x g_x - \sum_{\substack{x,y \\ x\neq y}} c_{xy} f_x g_y \quad (\text{since } c_x = \sum_y c_{xy} \text{ and } c_{xx} = 0) \\
&= -\sum_{x} a_{xx} f_x g_x - \sum_{\substack{x,y \\ x\neq y}} a_{xy} f_x g_y \\
&= -\sum_{x,y} a_{xy} f_x g_y.
\end{aligned}$$

□

7.8. **Cuts and Shorts.**

8. **Graphs and Random Walks** (after [Bar03])

Here we follow and expand a little on the summary in [Bar03].

[20] Think of the graph of $\mathcal{E}(f,f)$ as an $n-1$ dimensional parabola $\times$ straight line, sitting on $\mathbb{R}^n$, where n is the cardinality of V.

[21] It is the infinitesimal generator for the associated continuous time Markov process. ***Elaborate on this somewhere***

8.1. **Graphs.** Let
$$\Gamma = (G, E, a_{xy} \text{ for } \{x,y\} \in E),$$
be a finite or infinite *weighted graph* with set of vertices G and set of edges E. The weights satisfy
$$a_{xy} \ge 0, \quad a_{xy} > 0 \text{ iff } x \sim y.$$
where $x \sim y$ iff $\{x,y\}$ is an edge.

The *mass* at x is
$$\mu_0(x) := \sum_y a_{xy},$$
and this is extended to a measure μ_0 on G.

It is assumed $\mu_0(x) < \infty$ for all $x \in G$, which is clearly true if G is finite.

The *Laplacian* is defined by
$$\Delta f(x) = \sum_y \frac{a_{xy}}{\mu_0(x)}\big(f(y) - f(x)\big) = \left(\sum_y \frac{a_{xy}}{\mu_0(x)} f(x)\right) - f(x),$$
i.e. the difference between the weighted average of f at vertices y connected to x and the value of f at x.

The *Dirichlet form* is defined by
$$\begin{aligned}
\mathcal{E}(f,g) &= \frac{1}{2}\sum_{x,y} a_{xy}\big(f(x) - f(y)\big)\big(g(x) - g(y)\big) \quad \text{(sum of energy over all edges in case } f = g)\\
&= \sum_x \left(\sum_y a_{xy}\big(f(x) - f(y)\big)\right) g(x) \quad \text{(easy algebra)}\\
&= (-\Delta f, g)_{L^2(\mu_0)}.
\end{aligned}$$

8.2. **Random Walk and Markov Chain.** The *discrete time* random walk X_n is defined with transition probabilities from x to y given by
$$\mathbb{P}^x(y) = \frac{a_{xy}}{\mu_0(x)}.$$
We also write
$$P = (P_{xy}), \text{ where } P_{xy} = \mathbb{P}^x(y),$$
for the corresponding probability matrix. So
$$(Pf)(x) = \mathbb{E}^x f(X_1).$$
Note that
$$\Delta = P - I \text{ (i.e. } \Delta f = (P - I)f \text{ with } f \text{ a column vector)}; \quad \Delta f(x) = \mathbb{E}^x f(X_1) - f(x).$$

The *continuous time* random walk Y_t has jump times given by a Poisson process with parameter λ, where usually $\lambda = 1$. (Thus the expected number of jumps in a unit time interval is λ.) When at x the process jumps to each neighbour vertex y with probability $\mathbb{P}^x(y)$.

Thus, with P^n denoting the nth power of P,
$$\begin{aligned}
\mathbb{P}^x(X_n = y) &= (P^n)_{xy} =: P^n(x,y),\\
\mathbb{P}^x(Y_t = y) &= \sum_{n \ge 0} e^{-\lambda t}\frac{(\lambda t)^n}{n!}(P^n)_{xy} =: Q_t(x,y),
\end{aligned}$$
since the probability of exactly n jumps in $[0,t]$ is $e^{-\lambda t}(\lambda t)^n/n!$.

Note that
$$Q_t = \sum_{n\ge 0} e^{-\lambda t}\frac{(\lambda t)^n}{n!}P^n = e^{-\lambda t + \lambda t P} = e^{\lambda t\,\Delta}, \quad (Q_t f)(x) = \mathbb{E}^x f(Y_t).$$
So Δ (if $\lambda = 1$) is the infinitesimal generator of Y.

The transition matrix P is not symmetric. It is more convenient to work with the *transition density* or *heat kernel* for X:

$$p(x,y) := \frac{\mathbb{P}^x(y)}{\mu_0(y)} = \frac{a_{xy}}{\mu_0(x)\,\mu_0(y)},$$

which clearly is symmetric. More generally

$$p_0(x,y) = \frac{\delta(x,y)}{\mu_0(y)}, \quad p_n(x,y) := \frac{\mathbb{P}^x(X_n = y)}{\mu_0(y)} = \frac{\sum_{x_1,\dots,x_{n-1}} a_{xx_1}\cdot a_{x_1x_2}\cdot\dots\cdot a_{x_{n-1}y}}{\mu_0(x)\,\mu_0(y)},$$

which is again symmetric. Using $\Delta = P - I$, or directly, it is straightforward to check that p_n satisfies the *discrete heat equation*

$$p_0(x,y) = \frac{\delta(x,y)}{\mu_0(y)}, \quad p_{n+1}(x_0,x) - p_n(x_0,x) = \Delta_x\, p_n(x_0,x), \quad n \ge 0.$$

Similarly define the *transition density* or *heat kernel* for Y:

$$q_0(x,y) := \frac{\delta(x,y)}{\mu_0(y)} = p_0(x,y), \quad q_t(x,y) := \frac{\mathbb{P}^x(Y_t = y)}{\mu_0(y)} = \sum_{n\ge 0} e^{-\lambda t}\frac{(\lambda t)^n}{n!}\, p_n(x,y) \text{ if } t > 0,$$

which is again symmetric. It follows that q_t satisfies the heat equation

$$q_0(x,y) := \frac{\delta(x,y)}{\mu_0(y)}, \quad \frac{\partial q_t}{\partial t}(x_0,x) = \Delta_x\, q_t(x_0,x).$$

Estimates for p_n transfer easily to q_t.

9. Matrices

9.1. **Notation.** Consider $G \subset V$ and let $H = V \setminus G$.

Typically one has a sequence of vertices $(V_n)_{n\ge 0}$ of prefractal approximations to a fractal. One takes $V = V_n$ and either $G = V_{n-1}$, or V_0, or a for some $a \in V_0$, or some subset of V_0.

For $f : V \to \mathbb{R}$ let f_G and f_H denote the restrictions to G and H respectively.

With A as in (49) write

$$A = \begin{bmatrix} A_{GG} & A_{GH} \\ A_{HG} & A_{HH} \end{bmatrix}, \tag{51}$$

with an obvious notation.

Let D be the *diagonal matrix* $D_{xy} = c_x \delta_{xy}$. Let D_G and D_H be the restrictions of D to G and H respectively.

Let P be the standard *transition matrix* $P_{xy} = c_{xy}/c_x$.

Then the *Laplacian* is

$$\boxed{\Delta = P - I.} \tag{52}$$

It follows[22]

$$\boxed{A = D(P - I) = D\,\Delta\,.} \tag{53}$$

Note that A is symmetric, but Δ and P are not.

9.2. **Harmonic Extensions.**

Proposition 9.1. *Suppose $f_G = g$ is prescribed and $f_H = h$ is the extension to H such that f is harmonic on H. Then*

$$\boxed{h = -A_{HH}^{-1} A_{HG}\, g.} \tag{54}$$

In this case,

$$\boxed{\mathcal{E}(f) = -g^T(A_{GG} - A_{GH} A_{HH}^{-1} A_{HG})g.} \tag{55}$$

[22]Note that for any matrix B with the same number of rows as D, DB is obtained from B by multiplying all terms in row i by D_{ii}. Similarly, if B has the same number of columns as D then BD is obtained from B by multiplying all terms in column j by D_{jj}.

Proof. Proceed in the standard manner. [23] (Alternatively, since we know that the minimiser is harmonic we can also argue directly as in the Remark following the proof.)

$$
\begin{aligned}
-\mathcal{E}(f) &= \begin{bmatrix} g^T & h^T \end{bmatrix} \begin{bmatrix} A_{GG} & A_{GH} \\ A_{HG} & A_{HH} \end{bmatrix} \begin{bmatrix} g \\ h \end{bmatrix} \\
&= g^T A_{GG} g + g^T A_{GH} h + h^T A_{HG} g + h^T A_{HH} h \quad \text{(sum of scalars)} \\
&= g^T A_{GG} g + 2g^T A_{GH} h + h^T A_{HH} h \quad \text{(take transpose of previous third term).} \qquad (56)
\end{aligned}
$$

Since the minimum of $\mathcal{E}(f)$ occurs at h, replace h by $h + t\phi$, differentiate w.r.t. t, and set $t = 0$ to get, $\forall \phi : H \to \mathbb{R}$,

$$0 = 2g^T A_{GH} \phi + h^T A_{HH} \phi + \phi^T A_{HH} h = 2g^T A_{GH} \phi + 2h^T A_{HH} \phi.$$

It follows $g^T A_{GH} + h^T A_{HH} = 0$, so $A_{HG} g + A_{HH} h = 0$, and so

$$h = -A_{HH}^{-1} A_{HG} g.$$

Substituting into (56) gives

$$
\begin{aligned}
-\mathcal{E}(f) &= g^T (A_{GG} - 2g^T A_{GH} A_{HH}^{-1} A_{HG} + A_{GH} A_{HH}^{-1} A_{HH} A_{HH}^{-1} A_{HG}) g \\
&= g^T (A_{GG} - A_{GH} A_{HH}^{-1} A_{HG}) g.
\end{aligned}
$$

□

Remark: We can also obtain (54) buy using the fact h is harmonic. That is, from (53) the equations for the harmonic extension $f = \begin{bmatrix} g \\ h \end{bmatrix}$ of g can be written

$$\begin{bmatrix} I_G & O_{GH} \\ D_H^{-1} A_{HG} & D_H^{-1} A_{HH} \end{bmatrix} \begin{bmatrix} g \\ h \end{bmatrix} = \begin{bmatrix} g \\ 0 \end{bmatrix}$$

Hence

$$
\begin{aligned}
D_H^{-1}(A_{HG} g + A_{HH} h) &= 0 \\
\therefore\ h &= -A_{HH}^{-1} A_{HG}\, g.
\end{aligned}
$$

9.3. **Green's Function.**

Proposition 9.2. *Let $\mathcal{G}_a^G(x)$ be the Green function as in Section 5.1. Let $\mathcal{G}^G$ be the square matrix with column a denoted by $\mathcal{G}_a^G$ and whose entries are $\mathcal{G}_a^G(x)$ for $x \in H$.*

Then

$$\boxed{\mathcal{G} = -A_{HH}^{-1}.} \qquad (57)$$

Proof.

$$
\begin{aligned}
(I_H - P_{HH}) \mathcal{G}_a &= \frac{1}{c_a} \mathcal{X}_a \quad \text{(from (52) and (25))} \\
\therefore (I_H - P_{HH}) \mathcal{G} &= D_H^{-1} \quad \text{(by definition of } D\text{)} \\
\therefore \mathcal{G} &= (I_H - P_{HH})^{-1} D_H^{-1} = -A_{HH}^{-1} \quad \text{(by (53)).}
\end{aligned}
$$

□

9.4. **Other Quantities.**

(1) *Resistance Metric*: Let $G = \{y\}$. Then from (30), $\mathbb{R}_{xy} = \mathcal{G}_x^{\{y\}}(x)$, and so from (57),

$$\boxed{\mathbb{R}_{xy} = -(A_{HH}^{-1})_{xx} \text{ if } G = \{y\}.} \qquad (58)$$

(2) *Visit Times*: Let $\widetilde{P}$ be the transition matrix for the RW with *absorbing states* G. Then clearly $-D^{-1} A_{HH} = I_{HH} - \widetilde{P}$ and so

$$-A_{HH}^{-1} D = (I_{HH} - \widetilde{P})^{-1} = I_{HH} + \widetilde{P} + \widetilde{P}^2 + \cdots + \widetilde{P}^k + \cdots$$

Since $\widetilde{P}_{xy}^k = P^x(\text{RW is at } y \text{ after } k \text{ steps})$, the xy element of the right side is $\mathbb{E}^x V_{y \to G}$. Hence

$$\boxed{\mathbb{E}^x V_{y \to G} = -c_y (A_{HH}^{-1})_{xy}.} \qquad (59)$$

[23] Consider the minimum of $ag^2 + 2bgh + ch^2$ for a fixed scalar g and variable scalar h. The derivative is zero at $2bg + 2ch = 0$. Substituting $h = -c^{-1} b$ the minimum value is $(a - b^2 c^{-1}) g^2$.

(3) *Crossing Times*: Since $\mathbb{E}^x T_G = \sum_y \mathbb{E}^x V_{y\to G}$,

$$\boxed{\mathbb{E}^x T_G = -\sum_y c_y (A_{HH}^{-1})_{xy}.} \tag{60}$$

10. Matrix Examples

All computations done in Maple. See accompanying worksheet.

10.1. Simple Triangle T.

Consider the triangle network $\{1, 2, 3\}$ with unit resistances in Figure 5.

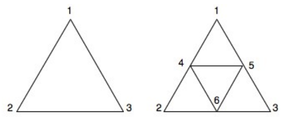

FIGURE 5. The triangle T and the $SG(2)$ prefractal

Then

$$A = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$

With $G = \{1\}$,

$$-A_{HH}^{-1} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$$

Going down the diagonal, the effective resistances are

$$\mathbb{R}_{12} = \mathbb{R}_{13} = 2/3.$$

The expected crossing times from x to 1 are obtained by multiplying the elements of column x by the degrees of the elements and summing:

$$\mathbb{E}^2 T_1 = \mathbb{E}^3 T_1 = 2.$$

With $G = \{1, 2\}$,

$$-A_{HH}^{-1} = \frac{1}{2}.$$

So

$$\mathbb{E}_{\{1,2\}3} = \frac{1}{2},\ \mathbb{E}^3 T_{\{1,2\}} = 2 \times \frac{1}{2} = 1,$$

as is clear anyway.

The crossing time theorem $\mathbb{E}^1 T_2 + \mathbb{E}^2 T_1 = \mathbb{R}_{12}\, C(\mathbb{G}) = 4$ is satisfied.

10.2. SG(2) Prefractal.

The network is shown in Figure 5. Each of the 6 edges is given unit resistance.

Then

$$A = \begin{bmatrix} -2 & 0 & 0 & 1 & 1 & 0 \\ 0 & -2 & 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 0 & 1 & 1 \\ 1 & 1 & 0 & -4 & 1 & 1 \\ 1 & 0 & 1 & 1 & -4 & 1 \\ 0 & 1 & 1 & 1 & 1 & -4 \end{bmatrix}, \tag{61}$$

We consider the three cases $G = \{1\}, \{1,2\}, \{1,2,3\}$. Then one computes respectively:

(62)
$$-A_{HH}^{-1} = \begin{bmatrix} \frac{10}{9} & 5/9 & 5/9 & 4/9 & 2/3 \\ 5/9 & \frac{10}{9} & 4/9 & 5/9 & 2/3 \\ 5/9 & 4/9 & \frac{11}{18} & \frac{7}{18} & 1/2 \\ 4/9 & 5/9 & \frac{7}{18} & \frac{11}{18} & 1/2 \\ 2/3 & 2/3 & 1/2 & 1/2 & 5/6 \end{bmatrix},$$
$$\begin{bmatrix} 5/6 & 1/6 & 1/3 & 1/3 \\ 1/6 & 1/3 & 1/6 & 1/6 \\ 1/3 & 1/6 & \frac{13}{30} & \frac{7}{30} \\ 1/3 & 1/6 & \frac{7}{30} & \frac{13}{30} \end{bmatrix}, \qquad \begin{bmatrix} 3/10 & 1/10 & 1/10 \\ 1/10 & 3/10 & 1/10 \\ 1/10 & 1/10 & 3/10 \end{bmatrix}.$$

In particular
$$\begin{aligned} \mathbb{R}_{12} = \mathbb{R}_{13} &= \frac{10}{9} \text{ (first matrix, first 2 diagonal elements)} \\ \mathbb{E}^2 T_1 = \mathbb{E}^3 T_1 &= 10 \text{ (first matrix, first two columns} \times \text{corresponding conductances)} \\ \mathbb{R}^3_{\{1,2\}} &= \frac{5}{6} \text{ (second matrix, first diagonal element)} \\ \mathbb{E}^3 T_{\{1,2\}} &= 5 \text{ (second matrix, first column} \times \text{corresponding conductances)} \\ C\mathbb{G}) &= 3 \times 2 + 3 \times 4 = 18. \end{aligned}$$

The crossing time theorem $\mathbb{E}^2 T_1 + \mathbb{E}^1 T_2 = \mathbb{R}_{12}\, C(\mathbb{G}) = 20$ is again satisfied.

Mass, Time and Resistance Scaling: Since edges all have unit resistance, i.e. the network is simple, the mass is $\sum_x \text{degree}(x)$, or equivalently twice the number of edges, or equivalently 6 times the number of triangles, i.e. 18. But the mass of T is 6. So the mass scaling m is $18/6 = 3$.

Next compare the crossing times and effective resistances above with those for T on page 21. Let τ and ρ, respectively, denote the factors by which the walk times and resistance between comparable nodes scale up in passing from T to $SG(2)$.

Then $m = 18/6 = 3$, $\tau = 10/2$ or $5/1 = 5$, and $\rho = \frac{10}{9}\Big/\frac{2}{3}$ or $\frac{5}{6}\Big/\frac{1}{2} = \frac{5}{3}$. That is:

(63)
$$\boxed{m = 3,\ \tau = 5,\ \rho = 5/3.}$$

Note the Einstein relation $\tau = m \times \rho$ holds.

10.3. **SG(3) Prefractal.** The network is shown in Figure 6. Each of the 18 edges is given unit resistance.

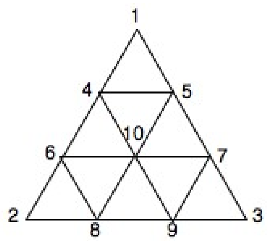

FIGURE 6. The $SG(3)$ prefractal

We consider the cases $G = \{1\}, \{1,2\}$. Then one computes:

$$A = \begin{bmatrix} -2 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -4 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & -4 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & -4 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & -4 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -4 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & -4 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & -6 \end{bmatrix}, \tag{64}$$

and

$$A_{HH}^{-1} = \begin{bmatrix} \frac{10}{7} & 5/7 & \frac{11}{21} & \frac{10}{21} & \frac{19}{21} & 2/3 & \frac{20}{21} & \frac{16}{21} & 5/7 \\ 5/7 & \frac{10}{7} & \frac{10}{21} & \frac{11}{21} & 2/3 & \frac{19}{21} & \frac{16}{21} & \frac{20}{21} & 5/7 \\ \frac{11}{21} & \frac{10}{21} & \frac{17}{28} & \frac{11}{28} & \frac{15}{28} & \frac{13}{28} & \frac{43}{84} & \frac{41}{84} & 1/2 \\ \frac{10}{21} & \frac{11}{21} & \frac{11}{28} & \frac{17}{28} & \frac{13}{28} & \frac{15}{28} & \frac{41}{84} & \frac{43}{84} & 1/2 \\ \frac{19}{21} & 2/3 & \frac{15}{28} & \frac{13}{28} & \frac{83}{84} & \frac{53}{84} & \frac{23}{28} & \frac{59}{84} & \frac{29}{42} \\ 2/3 & \frac{19}{21} & \frac{13}{28} & \frac{15}{28} & \frac{53}{84} & \frac{83}{84} & \frac{59}{84} & \frac{23}{28} & \frac{29}{42} \\ \frac{20}{21} & \frac{16}{21} & \frac{43}{84} & \frac{41}{84} & \frac{23}{28} & \frac{59}{84} & \frac{13}{12} & \frac{23}{28} & \frac{31}{42} \\ \frac{16}{21} & \frac{20}{21} & \frac{41}{84} & \frac{43}{84} & \frac{59}{84} & \frac{23}{28} & \frac{23}{28} & \frac{13}{12} & \frac{31}{42} \\ 5/7 & 5/7 & 1/2 & 1/2 & \frac{29}{42} & \frac{29}{42} & \frac{31}{42} & \frac{31}{42} & \frac{17}{21} \end{bmatrix},$$

$$\begin{bmatrix} \frac{15}{14} & 3/14 & 2/7 & 3/14 & 4/7 & 2/7 & 4/7 & \frac{5}{14} \\ 3/14 & \frac{523}{1260} & \frac{55}{252} & \frac{257}{1260} & \frac{277}{1260} & \frac{41}{252} & \frac{263}{1260} & \frac{5}{21} \\ 2/7 & \frac{55}{252} & \frac{113}{252} & \frac{41}{252} & \frac{79}{252} & \frac{43}{252} & \frac{65}{252} & \frac{11}{42} \\ 3/14 & \frac{257}{1260} & \frac{41}{252} & \frac{523}{1260} & \frac{263}{1260} & \frac{55}{252} & \frac{277}{1260} & \frac{5}{21} \\ 4/7 & \frac{277}{1260} & \frac{79}{252} & \frac{263}{1260} & \frac{853}{1260} & \frac{65}{252} & \frac{587}{1260} & \frac{5}{14} \\ 2/7 & \frac{41}{252} & \frac{43}{252} & \frac{55}{252} & \frac{65}{252} & \frac{113}{252} & \frac{79}{252} & \frac{11}{42} \\ 4/7 & \frac{263}{1260} & \frac{65}{252} & \frac{277}{1260} & \frac{587}{1260} & \frac{79}{252} & \frac{853}{1260} & \frac{5}{14} \\ \frac{5}{14} & \frac{5}{21} & \frac{11}{42} & \frac{5}{21} & \frac{5}{14} & \frac{11}{42} & \frac{5}{14} & \frac{19}{42} \end{bmatrix},$$

respectively.

Multiplying by the conductance vectors [2 2 4 4 4 4 4 4 6] and [2 4 4 4 4 4 4 6] respectively, we get the expected transition time vectors
$\begin{bmatrix} \frac{180}{7} & \frac{180}{7} & 17 & 17 & \frac{167}{7} & \frac{167}{7} & \frac{179}{7} & \frac{179}{7} & \frac{162}{7} \end{bmatrix}$ and $\begin{bmatrix} \frac{90}{7} & \frac{53}{7} & \frac{59}{7} & \frac{53}{7} & \frac{83}{7} & \frac{59}{7} & \frac{83}{7} & \frac{72}{7} \end{bmatrix}$.

In particular

$$\mathbb{R}_{12} = \mathbb{R}_{13} = \frac{10}{7} \text{ (first } A_{HH}^{-1}\text{, first 2 diagonal elements)}$$

$$\mathbb{E}^2 T_1 = \mathbb{E}^3 T_1 = 180/7 \text{ (first 2 entries of first transition vector)}$$

$$\mathbb{R}^3_{\{1,2\}} = \frac{15}{14} \text{ (second } A_{HH}^{-1}\text{, first diagonal element)}$$

$$\mathbb{E}^3 T_{\{1,2\}} = \frac{90}{7} \text{ (first entry of second transition vector)}$$

$$C(\mathbb{G}) = 3 \times 2 + 6 \times 4 + 6 = 36. \text{ (or } 2 \times 18\text{, each edge counted twice)}$$

The crossing time theorem $\mathbb{E}^2 T_1 + \mathbb{E}^1 T_2 = \mathbb{R}_{12}\, C(\mathbb{G}) = 360/7$ is again satisfied.

Mass, Time and Resistance Scaling: (Compare with $SG(2)$ for notation.) For $SG(3)$ we obtain the following mass, time and resistance scalings: $m = 36/6 = 6$, $\tau = \frac{180}{7}\Big/2$ or $\frac{90}{7}\Big/1 = \frac{90}{7}$ and $\rho = \frac{10}{7}\Big/\frac{2}{3}$ or $\frac{15}{14}\Big/\frac{1}{2} = \frac{15}{7}$. That is:

$$\boxed{m = 6,\ \tau = 90/7,\ \rho = 15/7.} \tag{65}$$

Note the Einstein relation $\tau = m \times \rho$ holds.

11. Trace Networks

11.1. **Motivation.** Consider the usual sequence of prefractal approximations to $SG(2)$ or $SG(3)$. Write this as

$$G_0 \subset G_1 \subset G_2 \subset \cdots \subset G_* := \bigcup_{n \geq 0} G_n.$$

Beginning with conductances 1 on the three edges of G_0 we want conductances on the edges of each G_n which give a sequence of compatible networks. That is, we require each G_n to be compatible with G_{n-1}, which means that if G_n is restricted to V_{n-1} it should give G_{n-1} in the black box sense of electrical networks, see Section 11.2. In fact, the conductance on each edge of G_n must be ρ^n, where $\rho = 5/3$ or $15/7$ for $SG(2)$ or $SG(3)$ respectively, and is the resistance scaling factor on page 22 and 24.

To see this, suppose initially that both G_0 and G_1 have unit resistance on each edge. Then from the discussion at the end of Sections 10.2 and 10.3 the effective resistances on V_0 *induced* from G_1 are $\rho\,\times$ the corresponding effective resistance on G_0. So to keep the same effective resistances on V_0 we need the resistance on each edge of G_1 to be ρ^{-1} and hence the conductance to be ρ. In a similar manner, the conductance on each edge of G_n should be ρ^n.

Note if conductances are multiplied by ρ then resistances are multiplied by ρ^{-1} and masses (obtained from the measure (c_x)) are multiplied by ρ. In particular the Einstein relation "time" = "mass" × "resistance" is preserved.

In general, suppose (V, A) is a network and $G \subset V$. Is there a network on G, i.e. is there a network (G, B), such that (V, A) and (G, B) have the same input/output response for sets of vertices from G? In other words, is there a network (G, B) which is indistinguishable from (V, A) in the black box sense if only vertices in G are accessible?

More precisely, is there a network (G, B) such that the following is true?

(66) *Suppose voltages are applied to some of the vertices in G and the corresponding currents into the circuit are measured. Then these currents are the same for both networks (V, A) and (G, B).*

(Recall that at any vertex where no voltage is applied the induced voltage is harmonic and so the current flows into the network.)

11.2. **Black Boxes.** Suppose we can apply arbitrary voltages at the various nodes in (V, A) and can then measure the current in or out of these nodes. (Note that if no voltage v is applied at x then $\Delta\, v(x) = 0$ and so the current into the network at x is zero.) Thus we treat the network as a "black box" for which we cannot directly access the wiring. In particular, we cannot directly measure the current across individual edges.

However, this "input-output" information is sufficient to reconstruct the network, that is to find the conductances c_{xy}.

(67) *Apply voltage* 1 *at x and* 0 *at all other nodes. Then a_{xy} is the flow out of the network through y* (an output quantity which can be measured in the black box sense).

To see this first note that for $x \neq y$ the current flow along xy is clearly $a_{xy} = c_{xy}$. But this is also the flow out of the network through y since there is no current flow between y and any node other than x. By conservation of current (4) the current *into* the network through x is $-a_{xx} = \sum_{\{y : y \neq x\}} a_{xy}$.

Conversely, once we know the a_{xy} we can find i_{xy} for arbitrary applied voltages by measuring voltage differences and using Ohm's law, and hence find the flow out of each vertex.

Note that finding the a_{xy} is different from finding the *effective* conductances $\mathbb{C}_{xy}$. In the latter case one applies voltage 1 at x and 0 at y, but no voltages are applied elsewhere. The current flow into the network through x or equivalently out of the network through y, is $\mathbb{C}_{xy}$.

Note also that the energy functional completely determines the conductances (and is trivially determined by the conductances), since for $x \neq y$, $a_{xy} = -\mathcal{E}(\delta_x, \delta_y)$ from (50).

Finally, the energy form determines the effective conductances, and the converse is true by induction on the number of vertices. (This is a bit of a calculation, see [Ki, p47]. Is there a better way?)

To summarise, one has the following information content equivalences:

(68) *input-output info.* $\Longleftrightarrow$ *conductances* $\Longleftrightarrow$ *Dirichlet form* $\Longleftrightarrow$ *effective conductances.*

11.3. **Trace Dirichlet Form.** *If* there exists a trace network (G, B) as in (66) then the corresponding Dirichlet form $\mathcal{E}_B$ must be given by $\mathcal{E}_B(g) = \mathcal{E}_A(\widetilde{g})$ for $g: G \to \mathbb{R}$, where $\widetilde{g}$ is the harmonic extension of g to V. This follows since

$$\begin{aligned}
\mathcal{E}_B(g) &= \sum_{x\in G} g(x)\, i^g(x) \quad \text{from (42)}\\
&= \sum_{x\in G} \widetilde{g}(x)\, i^{\widetilde{g}}(x) \quad \text{since } \widetilde{g} \text{ extends } g \text{ and from (66)}\\
&= \mathcal{E}_A(\widetilde{g}) \quad \text{from (42)}.
\end{aligned}$$

Motivated by this we make the following definition. The last equality in (69) is from (43).

Definition 11.1. Suppose (V, A) is a network. The *trace form* on $G \subset V$ is given by

$$\boxed{\mathcal{E}_B(g) = \mathrm{Tr}(\mathcal{E}|G)(g) := \mathcal{E}_A(\widetilde{g}) = \min\{\mathcal{E}_A(f) \mid f: V \to \mathbb{R},\ f = g \text{ on } G\},} \tag{69}$$

for any $g: G \to \mathbb{R}$, where $\widetilde{g}$ is the unique harmonic extension of g to V.

Denote the *trace network* by

$$\boxed{\mathrm{Tr}(A \mid G).} \tag{70}$$

Since the form determines the network and conversely, we occasionally use the two notations interchangeably.

11.4. **Construction of Trace Network.** Since $\widetilde{g}$ is linear in g it is clear from (69) that one can write

$$\mathcal{E}_B(g) = -\sum_{x,y\in G} b_{xy} g_x g_y$$

where $B = (b_{xy})$ is a symmetric bilinear matrix and $-B$ is non negative definite, c.f. (55).

Proposition 11.2. *The matrix $B = (b_{xy})$ defines a network (G, B). That is*

$$b_{xy} = b_{yx} \geq 0 \text{ if } x \neq y, \quad \sum_y b_{xy} = 0\ \forall x \in G.$$

Moreover, (66) is true.

In terms of matrices,

$$\boxed{B = A_{GG} - A_{GH} A_{HH}^{-1} A_{HG}.} \tag{71}$$

Proof. Since B is symmetric, $b_{xy} = b_{yx}$.

We claim that for $x \neq y$, b_{xy} is the flow out of V through x when a unit voltage is applied at y, zero voltage is applied elsewhere on G, and no voltages are applied in $V \setminus G$.

To see this note

$$\begin{aligned}
-\sum_{x,y\in G} b_{xy} f(x) g(y) &= \mathcal{E}_B(f, g) \quad \text{by definition of } b_{xy}\\
&= \mathcal{E}_A(\widetilde{f}, \widetilde{g}) \quad \text{by definition of } \mathcal{E}_B\\
&= \sum_{x\in G} \widetilde{f}(x) i^{\widetilde{g}}(x) \quad \text{from (42)}.
\end{aligned}$$

Let $f = \mathcal{X}_{\{x\}}$ and $g = \mathcal{X}_{\{y\}}$. The fact $i^{\widetilde{g}}(x) \le 0$ is clear from the current interpretation. Analytically it follows from the maximum principle applied at x and recalling the current is the negative of the Laplacian times the conductance weighting.

The fact $\sum_y b_{xy} = 0$ is conservation of current, see (7).

*** The map from voltages to currents is essentially the Laplacian and is really just B itself

$$i^g = Bg.$$

Hence the terminology "harmonic operator". This is another way of looking at (42). The result follows from this as in the beginning of the proof. (This needs cleaning up) ***

The matrix representation is immediate from (55). □

Example 11.3. Let V^1 be the network given by the $SG(2)$ prefractal in Figure 5 in which each of the 9 edges has conductance 1. Computing B_{GG} with A as in Section 10.3 gives the trace network on $V^0 = \{1,2,3\}$ in which each of the 3 edges has conductance 3/5. By an essentially trivial scaling argument, if the 9 edges in V^1 are each given conductance 5/3 then it follows that the conductances of the trace network on V^0 are 1.

Similarly, if V^1 is the network given by the $SG(3)$ prefractal in Figure 6 in which each of the 18 edges has conductance 1, then the trace network on $V^0 = \{1,2,3\}$ gives each of the 3 edges the conductance 7/15. Hence if the initial edges in V^1 are given conductance 15/7 then the conductances of the trace network on V^0 are 1.

11.5. **Tower Property of Trace.**

Proposition 11.4. *Suppose (V,A) is a network and $H \subset G \subset V$. Suppose (G,B) is the trace of (V,A) on G, (H,C) is the trace of (G,B) on H, and $(H,\widetilde{C})$ is the trace of (V,A) on H. Then*

$$\mathcal{E}_{\widetilde{C}} = \mathcal{E}_C, \quad (H,\widetilde{C}) = (H,C); \quad i.e. \ \boxed{\mathrm{Tr}\big(\mathrm{Tr}(A \mid G) \mid H\big) = \mathrm{Tr}(A \mid H).} \tag{72}$$

Suppose $h : H \to \mathbb{R}$, g is the harmonic extension of h to (G,B), f is the harmonic extension of g to (V,A), and f^* is the harmonic extension of h to (V,A). Then $f = f^*$.

Proof. Note that the two assertions in (72) are equivalent in the sense that the network determines the Dirichlet form and conversely.

Since $f = f^* = h$ on H and f^* is the energy minimising restriction,

$$\mathcal{E}_A(f^*) \le \mathcal{E}_A(f). \tag{73}$$

On the other hand, setting $g^* = f^*|_B$,

$$\begin{aligned} \mathcal{E}_A(f) &= \mathcal{E}_B(g) \quad \text{by definition of } \mathcal{E}_B, \\ &\le \mathcal{E}_B(g^*) \end{aligned}$$

since $g^* = g = h$ on H and g is the energy minimising extension of h,

$$= \mathcal{E}_A(f^*)$$

since f^* is the energy minimising extension of g^* (being the energy minimising extension of the restriction h of g^*) and by definition of $\mathcal{E}_B$.

It follows

$$\mathcal{E}_A(f) = \mathcal{E}_A(f^*).$$

Hence $f = f^*$ as f^* is the *unique* energy minimising extension of h.

Finally

$$\mathcal{E}_{\widetilde{C}}(h) = \mathcal{E}_A(f^*) = \mathcal{E}_A(f) = \mathcal{E}_B(g) = \mathcal{E}_C(h),$$

where the first, third and fourth equalities are by the definition of $\mathcal{E}_{\widetilde{C}}$, $\mathcal{E}_B$ and $\mathcal{E}_C$. □

Example 11.5. The second paragraph in the Proposition is certainly not true if one works with a *restrictions* of a network, rather than with *traces* of a networks.

For example, consider a simple (i.e. unit conductances) network with vertices $V = \{1, 2, 3, 4\}$ and edges $E = \{12, 23, 34, 41\}$. Let $H = \{1, 2\}$ and $G = \{1, 2, 3\}$.

Then given $h : H \to \mathbb{R}$ and using the notation of the Proposition, one easily checks

$$g(3) = h(2), \quad f(4) = \frac{1}{2}(h(1) + g(3)) = \frac{1}{2}(h(1) + h(2)), \quad f^*(4) = \frac{2}{3}h(1) + \frac{1}{3}h(2).$$

To compute the trace network on G let

$$A = \begin{bmatrix} -2 & 1 & 0 & 1 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 1 & 0 & 1 & -2 \end{bmatrix}$$

be the generator matrix for (V, A). Then from (71) the generator matrix for the trace network on G is

$$B = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} & 1 & \frac{1}{2} \\ 1 & -2 & 1 \\ \frac{1}{2} & 1 & -\frac{3}{2} \end{bmatrix}.$$

So the conductances on G are $b_{12} = b_{23} = 1$, $b_{13} = \frac{1}{2}$. It follows that the harmonic extension g to G of $h : H \to \mathbb{R}$ is given by $g(3) = \frac{1}{3}h(1) + \frac{2}{3}h(2)$. The harmonic extension f of g to V is then given by $f(4) = \frac{2}{3}h(1) + \frac{1}{3}h(2)$.

In this case $f = f^*$.

Example 11.6. Consider the prefractal example in Figure 5, where each edge is given conductance 1.

Using (71) one calculates the induced network on $\{2, 4, 6\}$ is given by

$$B = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -\frac{11}{4} & \frac{7}{4} \\ 1 & \frac{7}{4} & -\frac{11}{4} \end{bmatrix}.$$

The effective conductances of each edge using B can easily be calculated by hand as in Section 3.5 to be $18/11$ on edges 24 and 26, and $9/4$ on edge 46.

On the other hand, directly computing the effective resistances, for example as in (58), and inverting to get the conductances, gives the same result.

That is, taking the trace on $\{2, 4, 6\}$ and then taking its trace on one of its edge in $\{2, 4, 6\}$, is the same as taking the trace directly on the edge.

12. PCF fractals

12.1. **Overview.** Suppose $\psi_1, \dots, \psi_M$ are contractions on a complete metric space X with compact attractor K.

In certain situations which we later explain, roughly speaking for finitely ramified fractals, and more precisely for post criticially finite [pcf] fractals, we construct a metric on K, as well as a Dirichlet form, Laplacian and diffusion process, which do not depend on the metric on X or any imbedding of K into Euclidean space.

The construction will depend only on "topological information" about K and some "resistance structure information" concerning certain finite subsets of K. In particular, the construction does not depend on the imbedding of K in some Euclidean space.[24]

Topological Information: We construct a sequence of (finite for pcf fractals) subsets of K,

$$V^{0)} \subset V^1 \subset V^2 \subset \cdots,$$

whose union is dense in K. Moreover,

$$\psi_i(V^0) \subset V^1 \text{ for } i = 1, \dots, M, \quad V^1 = \bigcup_{i=1}^{M} \psi_i(V^0),$$

[24]This commonly made statement is a little misleading. It will often be natural, for example, to select "resistance structure information" that reflects symmetries in K, where the symmetries in turn follow from some metric on K.

All the topological information is encoded in the combinatorial information describing the *finite* maps $\psi_i|_{V^0} : V^0 \to V^1$, see Section 12.6. This information carries over to the V^m for $m \geq 2$ via the facts

$$\psi_i(V^m) \subset V^{(m+1)}, \quad V^{(m+1)} = \bigcup_{i=1}^{M} \psi_i(V^m).$$

Resistance Information: The set V^0 is given "resistance" information for each pair of nodes. This information can be interpreted as an electrical network, or perhaps as how much work is required to move from one node to another. There will be additional "resistance scaling" information $\rho = (\rho_1, \dots, \rho_M)$ with $\rho_m > 0$ which allows replication of M *normalised* copies of the network on V^0 to be joined together at certain nodes to give a network on V^1, whose trace ("restriction") on V^0 will give back the original network on V^0. See Section 13.3.

This then allows one to build up a sequence of compatible networks and corresponding random walks on the V^m for each m. It is then possible to pass to a limit and define a Dirichlet form, Laplacian and diffusion process on the fractal K.

It is not known if such resistance structure information always exists, see Section 13.4. But there are large classes of situations where it does.

Once the topology is fixed as discussed previously, the construction of the Dirichlet form, Laplacian and diffusion process depends only on the resistance structure, and not otherwise on the geometry of the fractal. *However*, there are often canonical resistance structures related to the geometry of K when K *is* imbedded in $\mathbb{R}^d$. This then gives deep information about the fractal K related to its geometric structure as a subset of $\mathbb{R}^d$.

12.2. **Words and Addresses.** As before, suppose $\psi_1, \dots, \psi_M$ are contractions on a complete metric space X with compact attractor K.

In order to describe the necessary topological information we work with the symbol space on $\{1, \dots, M\}$.

First define

$$W_0 = \{\emptyset\}, \quad W_n = \{1, \dots, M\}^n, \quad W^* = \bigcup_{n\geq 0} W_n, W = \{1, \dots, M\}^{\mathbb{N}}, \tag{74}$$

$$\text{if } w = w_1 \dots w_n \in W_n \text{ then } |w| = n, \tag{75}$$

$$\psi_w = \psi_{w_1} \circ \cdots \circ \psi_{w_n}, \quad E_w = \psi_w(E). \tag{76}$$

$$\sigma(w_1 w_2 w_3 \dots) = w_2 w_3 \dots, \quad i(w) = iw = iw_1 w_2, \dots. \tag{77}$$

Thus W^* is the set of finite words, W_n is the set of words of length n, and the length of w is denoted by $|w|$. Words will be interpreted as the address of certain "cells" in K. Elements of W are infinite words and will be interpreted as addresses of points in K.

The left shift operator is σ and $i(\cdot)$ is the operator of adjoining i to the beginning of a word. We will soon see that at the level of operating on K rather than W, the operator σ corresponds to the ψ_i^{-1} on their domains, and is multiply defined on subsets of the form $K_i \cap K_j$; while the operator $i(\cdot)$ corresponds to ψ_i.

Concatenation of words is denoted by juxtaposition. That is, vw is the word whose initial symbols are given by the finite word v followed by the finite or infinite word w.

If w is a finite or infinite word then $w|n$ *is the initial segment of* w *of length* n. If w is of length less than n then $w|n = w$.

A *metric* d is defined on W in one of various equivalent ways. For example,

$$d(w, v) = 2^{-n} \text{ where } n \text{ is the least integer such that } w|n \neq v|n.$$

12.3. **Self-Similar Structure.** The topological information discussed in Section 12.1 can be obtained from the self-similar structure defined as follows.

Definition 12.1. $(K, \psi_1, \dots, \psi_M)$ is a *self-similar structure* if K is a compact, metrizable topological space, and $\exists$ a continuous map $\pi : W \to K$ which is onto and such that

$$\pi \circ i = \psi_i \circ \pi. \tag{78}$$

The set K is called a *self-similar set.* If $x = \pi(w)$ then w is an *address* of x.

The set $\bigcap_{n\ge1} K_{w|n}$ is a singleton (by continuity) and its sole element is $\pi(w)$.

Note:

(1) It is not necessary to assume K is compact as this follows from the fact it is the image of the compact space W.
(2) Metrizability follows from the weaker assumption that K is Hausdorff, see [Kam00, p 118, Remark 2.2].

12.4. **Post Critical Set.** Suppose $(K, \psi_1, \dots, \psi_M)$ is a self-similar structure as in Definition 12.1. See Table 12.4 and the examples in Section 12.5.

Set	Nodes	Words	Symbols
Ramification/Critical	R	$\mathcal{C}$	$\square$
Boundary	V^0	$\mathcal{P}=\mathcal{P}^0$	$\bigcirc$
Level 1 Boundary	V^1	$\mathcal{P}^1$	$\bullet$
Level m Boundary	V^m	$\mathcal{P}^m$	
All Levels	V^*	$\mathcal{P}^*$	

TABLE 1. Important sets in the pcf setting. The "$\bullet$" in later Figures refers to $V^1 \setminus V^0$.

Remark 12.2. The idea in the following is that the set V^0 should include *all* "level 0" points which are the pull back of level m ramification points for some m. For example, ramification points at level one for $SG(2)$ and $SG(3)$ are enclosed by squares in Figure 7 and pull back to the 3 circled points comprising V^0. Level 2 ramification points are not shown, but clearly they do not lead to any further points in V^0.

In the left diagram in Figure 9 the 6 level one ramification points are enclosed by squares and these lead to the 6 circled points comprising V^0. Higher level ramification points do not lead to any further points in V^0.

In the right diagram in Figure 9 the situation is more complex. The 6 level one ramification points are enclosed by squares and these lead to 6 of the 9 circled points comprising V^0. These are firstly the top, bottom left and bottom right circles; and secondly the bottom edge circle one in from the right, the right edge circle one down from the top, and the left edge circle one up from the bottom.

The right edge circle one up from the bottom corresponds to a ramification point P at level two given by where the triangle 32 meets the triangle 43. Even though P is also a ramification point at level one, it leads to different points in V^0 for levels one and level two. More precisely, P has coordinates $4\dot{3}$ and $3\dot{2}\dot{1}$. Pulling back from level one gives $\dot{3} \in V^0$ and $\dot{2}\dot{1} \in V^0$, but pulling back from level two gives $\dot{3} \in V^0$ and $\dot{1}\dot{2} \in V^0$. It is not hard to see that higher level ramification points do not give any further points in V^0.

Remark 12.3. The reason one defines $\mathcal{P} = \bigcup_{n\ge1} \sigma^n\mathcal{C}$, and not just $\mathcal{P} = \sigma\mathcal{C}$, is as follows.

The network on V^1 is obtained by replicating copies of the network on V^0, and tracing gives the network on V^0. So we want $V^1 = \bigcup_{i=1}^M \psi_i V^0$ and $V^0 \subset V^1$. The first requirement implies at the address level that $w \in \mathcal{P}^1$ iff $\sigma w \in \mathcal{P}^0$. Then from the second requirement $w \in \mathcal{P}^0$ implies $\sigma w \in \mathcal{P}^0$, that is $\mathcal{P}$ is closed under σ.

Definition 12.4. The set of (level one) *ramification points* of K is the set of points in two or more "components" $\psi_i K$. The *critical set* $\mathcal{C}$ is the set of addresses of ramification points. The *post critical set* $\mathcal{P}$ is the set of "ancestral addresses" of addresses in $\mathcal{C}$.

$$\boxed{R = \bigcup_{i\ne j} K_i \cap K_j, \quad \mathcal{C} = \pi^{-1}(R), \quad \mathcal{P} = \bigcup_{n\ge1} \sigma^n\mathcal{C}.} \tag{79}$$

The *boundary* V^0 of K is the set of points with address in the postcritical set. The *set of m-vertices* or the set of m-level boundary points, and hence the *set of all vertices*, are defined from the boundary points by pre-concatenating words of length m.

$$\boxed{V^0 := \pi(\mathcal{P}), \quad V^m := \bigcup_{|w|=m} V^0_w = \bigcup_{i=1}^{M} \psi_i V^{m-1}, \quad V^* := \bigcup_{m\geq 1} V^m.} \tag{80}$$

It follows from the definitions that

$$R \subset V^1, \quad V^0 \subset V^1 \subset V^2 \subset \cdots, \tag{81}$$

$$\boxed{K_w \cap K_v = V^0_w \cap V^0_v}, \quad |w| = |v|, \ w \neq v. \tag{82}$$

The set of *addresses of m-vertices* is

$$\boxed{\mathcal{P}^m = \bigcup_{|w|=m} w\mathcal{P}}; \text{ it follows } \boxed{V^m = \pi\mathcal{P}^m, \quad \mathcal{P}^m = \pi^{-1}V^m}. \tag{83}$$

12.5. **Post Critically Finite Fractals.**

Definition 12.5. A self-similar structure $(K, \psi_1, \ldots, \psi_M)$ is *post critically finite* [pcf] if $\mathcal{P}$ defined in (79) is finite. We say K is a *pcf fractal set*.

A pcf fractal set is finitely ramified (in the physics terminology), i.e. R is finite. But a finitely ramifie fractal need not be pcf. See the last remark in Example 12.8 in this Section.

Define an equivalence relation on W by

$$v \sim w \text{ iff } \pi(v) = \pi(w). \tag{84}$$

Below are examples of pcf fractals and the associated sets. The map ψ_i in each case maps the largest triangle, square etc. onto the region i. See also Table 12.4.

Example 12.6 (Sierpinski gaskets). The situation here is particularly simple, in that $V^1 = V^0 \cup R$ and the union is disjoint. Moreover, the points in V^0 are fixed points of ψ_1, ψ_2, ψ_3.

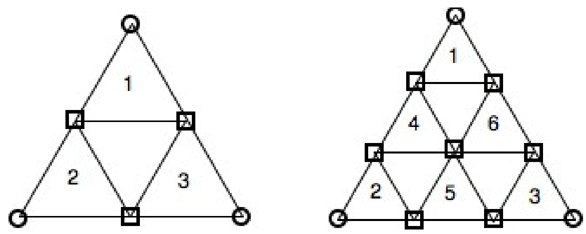

FIGURE 7. $SG(2)$ and $SG(3)$. See Table 12.4 for notation.

The relevant sets for $SG(2)$ are

$$\mathcal{C} = \{1\dot{2} \sim 2\dot{1}, 2\dot{3} \sim 3\dot{2}, 3\dot{1} \sim 1\dot{3}\}, \quad \mathcal{P} = \{\dot{1}, \dot{2}, \dot{3}\}, \quad \mathcal{P}^1 = \mathcal{C} \cup \mathcal{P},$$

and the union is disjoint.

For $SG(3)$ the relevant sets are

$$\mathcal{C} = \{1\dot{2} \sim 4\dot{1}, 4\dot{2} \sim 2\dot{1}, 2\dot{3} \sim 5\dot{2}, 5\dot{3} \sim 3\dot{2}, 3\dot{1} \sim 6\dot{3}, 6\dot{1} \sim 1\dot{3}, 4\dot{3} \sim 5\dot{1} \sim 6\dot{2}\},$$
$$\mathcal{P} = \{\dot{1}, \dot{2}, \dot{3}\}, \quad \mathcal{P}^1 = \mathcal{C} \cup \mathcal{P},$$

and the union is again disjoint.

Example 12.7 (Cut square). The diagram is misleading as it is not imbedded in $\mathbb{R}^2$, see [Bar98, pp 64,65].

Cuts are made from the centre of a unit square to each of the 4 edges along four intervals not including the endpoints. More precisely, points on these intervals are counted twice. A convenient metric is the geodesic metric.

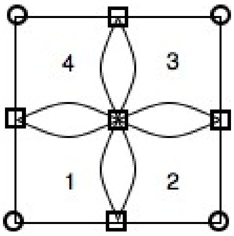

FIGURE 8. The cut square. See Table 12.4 for notation.

The relevant sets are

$$\mathcal{C} = \{1\dot{3} \sim 2\dot{4} \sim 3\dot{1} \sim 4\dot{2}, 1\dot{2} \sim 2\dot{1}, 2\dot{3} \sim 3\dot{2}, 3\dot{4} \sim 4\dot{3}, 4\dot{1} \sim 1\dot{4}\}, \quad \mathcal{P} = \{\dot{1}, \dot{2}, \dot{3}, \dot{4}\}, \quad \mathcal{P}^1 = \mathcal{C} \cup \mathcal{P},$$

and the union is again disjoint.

Example 12.8 ($SG(2)$ with added triangle). In this case V^0 is no longer the set of fixed points of some of the ψ_i. Moreover, in the first example $\mathcal{C} \cap \mathcal{P} \neq \emptyset$ and in both examples $\mathcal{C} \cup \mathcal{P} \subsetneq \mathcal{P}^1$.

The reason in each case is that the intersection $K_i \cap K_j$ is at different relative points in K_i and K_j.

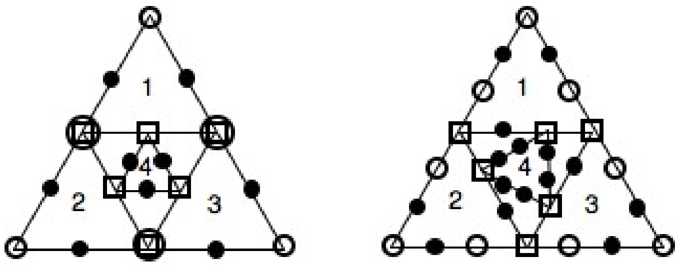

FIGURE 9. $SG(2)$ with added triangle. See Table 12.4 for notation.

In the first example the triangle 4 is equilateral and obtained from the large triangle by translation and scaling.

In the second case triangle 4 is again equilateral but its vertices are 2/3 along the relevant side of triangles 1, 2 or 3 in the clockwise direction. It is obtained from the original triangle by scaling, translating, and rotating $\pi/6$ in the clockwise direction.

The relevant sets in the first example are

$$\mathcal{C} = \{1\dot{2} \sim 2\dot{1}, 2\dot{3} \sim 3\dot{2}, 3\dot{1} \sim 1\dot{3}, 4\dot{1} \sim 12\dot{3} \sim 13\dot{2}, 4\dot{2} \sim 23\dot{1} \sim 21\dot{3}, 4\dot{3} \sim 31\dot{2} \sim 32\dot{1}\},$$

$$\mathcal{P} = \{\dot{1}, \dot{2}, \dot{3}, 1\dot{2} \sim 2\dot{1}, 2\dot{3} \sim 3\dot{2}, 3\dot{1} \sim 1\dot{3}\},$$

$$\mathcal{P}^1 = \mathcal{C} \cup \mathcal{P} \cup \{11\dot{2} \sim 12\dot{1}, 13\dot{1} \sim 11\dot{3}, 21\dot{2} \sim 22\dot{1}, 22\dot{3} \sim 23\dot{2}, 32\dot{3} \sim 33\dot{2}, 33\dot{1} \sim 31\dot{3},$$

$$41\dot{2} \sim 42\dot{1}, 42\dot{3} \sim 43\dot{2}, 43\dot{1} \sim 41\dot{3}\}.$$

The relevant sets in the second example are

$$\mathcal{C} = \{1\dot{2} \sim 2\dot{1}, 2\dot{3} \sim 3\dot{2}, 3\dot{1} \sim 1\dot{3}, 4\dot{1} \sim 13\dot{2}, 4\dot{2} \sim 21\dot{3}, 4\dot{3} \sim 32\dot{1}\},$$

$$\mathcal{P} = \{\dot{1}, \dot{2}, \dot{3}, \dot{1}\dot{2}, \dot{2}\dot{1}, \dot{2}\dot{3}, \dot{3}\dot{2}, \dot{3}\dot{1}, \dot{1}\dot{3}\}, \quad \mathcal{P}^1 = \mathcal{C} \cup \mathcal{P} \cup \{\text{“black dots”}\}.$$

By taking an irrational fraction of 2π for the rotation in the construction of triangle 4 it is straightforward to obtain examples where R is finite with cardinality 6, but $\mathcal{P}$ is infinite.

12.6. **Ancestral Systems.** (See [Kig93, Appendix A].) Since $\pi : W \to K$, where π is onto and continuous, W is compact and K is Hausdorff, there is a homeomorphism $W/\!\sim\, \cong K$, where "$\sim$" is the equivalence relation on W given as before by $v \sim w$ iff $\pi(v) = \pi(w)$. It follows that K is determined topologically by the information $(W, \sim)$.

But we will see that for pcf structures, $\sim$ and hence K up to homeomorphism, can be obtained from the *combinatorial information* given by the *finite* one-one maps $\psi|_{V^0} : V^0 \to V^1$ for $i = 1, \ldots, M$. This leads to the following definition.

Definition 12.9. $(U, V, G_1, \dots, G_M)$ is an *ancestral system* if $U \subset V$ are finite sets, each $G_i : U \to V$ is one-one, $V = \bigcup_{i=1}^M G_i(U)$ and $G_i(U) \setminus U \neq \emptyset$ if $U \neq \emptyset$.

In particular, it follows from the relevant definitions that if $(K, \psi_1, \dots, \psi_M)$ is a self-similar structure then $(V^0, V^1, \psi_1|_{V^0}, \dots, \psi_M|_{V^0})$ is an ancestral system. (The case $U = \emptyset$ in the definition corresponds to a totally disconnected fractal.)

Conversely, one can reconstruct a self-similar structure from an ancestral system. Moreover, up to the natural notions of isomorphism for ancestral systems and isomorphism for self-similar structures this correspondence is one-one.

To do this, the main point is to reconstruct the equivalence relation $\sim$ on W by using only the information in the ancestral system $(V^0, V^1, \psi_1|_{V^0}, \dots, \psi_M|_{V^0})$. By a similar construction an *arbitrary* ancestral system $(U, V, G_1, \dots, G_M)$ will give a self similar structure $(W/{\sim}, \psi_1, \dots, \psi_M)$, where $\psi_i \circ \pi = \pi \circ i$ and $\pi : W \to W/{\sim}$ is the usual projection map.

We use the fact that if $w \sim v$ and $w \neq v$ then there is a least n such that $w|n \neq v|n$. It follows

$$\pi(w) = \pi(v) \in \boxed{K_{w|n} \cap K_{v|n} = V^0_{w|n} \cap V^0_{v|n}} \subset V^n \subset V^*,$$

from (82). In particular, $w, v \in V^*$, and so in defining $\sim$ we need only consider points in $\mathcal{P}^* \subsetneq W$, as all other points belong to trivial singleton equivalence classes.

The key idea is as follows: For $w \neq v \in W$ set $w = u\hat{w}$ and $v = u\hat{v}$ where u is the maximum common initial segment. Then $w \sim v$ iff $\hat{w}$ and $\hat{v}$ are different addresses of the same point in V^1.

Moreover, the set $\mathcal{A}_x$ of all addresses of any point $x \in V^1$ can be obtained by chasing backwards into V^0 via the ψ_i^{-1}. That is, $w \in \mathcal{A}_x$ iff for some $(x_n)_{n \geq 1} \subset V^0$,

$$x = \psi_{w_1}(x_1) = \psi_{w_1 w_2}(x_2) = \psi_{w_1 w_2 w_3}(x_3) = \cdots .$$

For an arbitrary ancestral system $(U, V, G_1, \dots, G_M)$ we similarly define $\sim$. That is, for each $x \in V$ first define $w \in \mathcal{A}_x \subset W$ iff for some $(x_n)_{n \geq 1} \subset U$,

$$x = G_1(x_1) = G_1 G_2(x_2) = G_1 G_2 G_3(x_3) = \cdots .$$

Then define $w \sim v$ iff

(1) $w = v$, or
(2) $w = u\hat{w}$ and $v = u\hat{v}$ where, $u \in W_n$ for some $n \geq 0$, and $\hat{w}, \hat{v} \in \mathcal{A}_x$ for some $x \in V$.

13. Replication of a Network

13.1. **Notation.** Let $(K, \psi_1, \dots, \psi_M)$ be a pcf structure.

Let V^m be the set of level m-boundary points as in Section 12.4.

Let $(r_{xy})_{x,y \in V^0}$ be a set of resistances defining a network on V^0, where we allow $r_{xy} = +\infty$ as usual. Equivalently, let $A = (a_{xy})_{x,y \in V^0}$ be the conductance matrix. That is, $a_{xy} = r_{xy}^{-1}$ if $x \neq y$, and $a_{xx} = -\sum\{a_{xy} : y \neq x\}$ for fixed $x \in V^0$.

Let $\mathcal{E} = \mathcal{E}^0$ be the corresponding Dirichlet form on V^0. That is

$$\boxed{\mathcal{E}^0(f) = \mathcal{E}(f) = \frac{1}{2} \sum_{x,y \in V^0} a_{xy}\big(f(x) - f(y)\big)^2.} \tag{85}$$

Let $\rho = (\rho_1, \dots, \rho_M)$ be a *resistance scaling* vector of positive numbers. Typically $\rho_i > 1$.

The idea is that the resistance scaling factor ρ_i is the amount by which resistance distances are multiplied when going from the smaller K_i to the larger K. This is consistent with length scaling and mass scaling, both of which are greater than 1. Notice that ρ is also *conductance* scaling when passing in the *opposite* direction from K to K_i.[25]

For the IFS corresponding to $SG(2)$ the natural value is $\rho = (\frac{5}{3}, \frac{5}{3}, \frac{5}{3})$ and for $SG(3)$ it is $\rho = (\frac{15}{7}, \dots, \frac{15}{7})$. See Example 11.3 in Section 11.4.

In this and other cases where $\rho_1 = \cdots = \rho_M$ we also use ρ to denote the individual elements of the vector $(\rho_1, \dots, \rho_M)$.

[25] The resistance scaling vector ρ here is the same as the vector $\rho({r_1}^{-1}, \dots, {r_M}^{-1})$ in [Bar98, Section 6]. There, ρ is a *scalar* called the *resistance scaling factor* and $(r_1, \dots, r_M)$ is called the *resistance vector*. See [Bar98, p 92, Definition 6.26; p 79, before Definition 6.1].

For $f : V^n \to \mathbb{R}$ and $w \in W_n$, define

$$f_w = f \circ \psi_w : V^0 \to \mathbb{R}. \tag{86}$$

Thus f_w is essentially the restriction of f to the cell V^0_w, but interpreted as a function defined on the domain V^0 (which is independent of w).

13.2. **Definition of Replication.** We now use the ψ_i and ρ_i to make weighted copies of the network on V^0, which are then joined together at the ramification points $R = \bigcup_{i\neq j} V^0_i \cap V^0_j$ (see (79) and (82)) to give a network on V^1 (see (80)).

More precisely, the replicated Dirichlet form $\mathcal{E}^R$ and replicated conductance matrix $A^R = (a^R_{xy})$ on V^1 are defined by

$$\mathcal{E}^R(f) = \sum_{i=1}^{M} \rho_i \mathcal{E}(f_i), \quad \text{where } f : V^1 \to \mathbb{R}, \tag{87}$$

$$a^R_{xy} = \sum \{\rho_i a_{uv} : \psi_i(u) = x,\ \psi_i(v) = y\}. \tag{88}$$

(In the sum defining a^R_{xy} there may be more than one term. This occurs in the case of the cut square in Example 12.7.)

It follows from the definitions that A^R is a conductance matrix and that $\mathcal{E}^R$ is the associated Dirichlet form. That is,

$$\mathcal{E}^R(f) = \frac{1}{2} \sum_{x,y\in V^1} a^R_{xy}\big(f(x) - f(y)\big)^2. \tag{89}$$

13.3. **Invariance under Replication.**

Definition 13.1. Suppose $(K, \psi_1, \dots, \psi_M)$ is a pcf system, with sets V^0 and V^1, and resistance scaling vector $\rho = (\rho_1, \dots, \rho_M)$, as in Section 13.1. Then a network A on V^0 is *invariant under replication* (using ρ) followed by tracing if

$$\boxed{A = \mathrm{Tr}(A^R \mid V^0),} \tag{90}$$

where A^R is the replicated network on V^1.

It follows from Section 11.3 that

Proposition 13.2. *A is invariant under replication followed by tracing iff the replicated Dirichlet form $\mathcal{E}^R$ satisfies*

$$\mathcal{E}(f) = \mathcal{E}^R(\widetilde{f}) = \min\{\mathcal{E}^R(h) : h|_{V^0} = f\}, \quad \textit{i.e.}\ \boxed{\mathcal{E} = \mathrm{Tr}(\mathcal{E}^R \mid V^0)}. \tag{91}$$

where $\widetilde{f}$ is the (unique) harmonic extension of f to V^1.

Example 13.3. For $SG(2)$ the network A on V^0 with $a_{ij} = 1$ (or any fixed positive constant) for $i \neq j$ is invariant if $\rho = (\frac{5}{3}, \frac{5}{3}, \frac{5}{3})$. This follows from the discussion in Example 11.3.

Similarly, for $SG(3)$ the same network A is invariant if $\rho = (\frac{15}{7}, \frac{15}{7}, \frac{15}{7}, \frac{15}{7}, \frac{15}{7}, \frac{15}{7})$.

We usually just write $\rho = 5/3$ or $\rho = 15/7$ respectively.

13.4. **Finding Invariant Networks.** Motivated by Example 13.3 we have the following method.

General procedure: Suppose $(K, \psi_1, \dots, \psi_M)$ a pcf structure.

(1) Begin with a conductance matrix A on V^0 which reflects certain geometric or other features of K, following the idea in Examples 11.3 and 13.3,
(2) Select a putative resistance scaling vector $\widehat{\rho} = (\widehat{\rho}_1, \dots, \widehat{\rho}_M)$ which reflects the *relative* "resistance" sizes (after taking inverses of the $\widehat{\rho}_i$) of the V^0_i, and assemble the replicated conductance matrix $\widehat{A}^R$ on V^1 as in (88) or equivalently (87).
(3) Compute the trace conductance matrix $\widehat{B} = \mathrm{Tr}(\widehat{A}^R \mid V^0)$ on V^0 as in (71) or by the methods in Section 3.5.
(4) *Suppose*

$$\mathrm{Tr}(\widehat{A}^R \mid V^0) = \lambda A \quad \text{for some } \lambda. \tag{92}$$

It follows that the resistance scaling vector $\rho := \lambda^{-1}\widehat{\rho}$ gives the replicated conductance matrix $A^R = \lambda^{-1}\widehat{A}^R$ and the trace matrix $B = \lambda^{-1}\widehat{B}$. For this ρ, $\mathrm{Tr}(A^R \mid V^0) = A$ and so A *is invariant under replication using the resistance scaling vector* $\rho = \lambda^{-1}\widehat{\rho}$.

The eigenvalue λ for $\widehat{\rho}$ is used to give a new resistance scaling vector ρ for which the eigenvalue is 1. Thus $\rho = (\rho_1, \dots, \rho_M)$ reflects the *absolute* "resistance" sizes of the V_i^0.[26]

More General procedure:

(1) Begin with a class $\mathcal{A}$ of conductance matrices A on V^0 which reflect certain geometric or other features of K.
(2) Using a putative resistance scaling vector $\widehat{\rho} = (\widehat{\rho}_1, \dots, \widehat{\rho}_M)$ which reflects the *relative* "resistance" sizes of the V_i^0 but is *independent* of $A \in \mathcal{A}$, assemble the replicated conductance matrices $\widehat{A}^R$ on V^1 for $A \in \mathcal{A}$.
(3) Compute the trace conductance matrices $\widehat{B} = \mathrm{Tr}(\widehat{A}^R \mid V^0)$ on V^0 as in (71) or by the methods in Section 3.5.
(4) Compute those $A \in \mathcal{A}$ for which there exists $\lambda > 0$ such that

$$\mathrm{Tr}(\widehat{A}^R \mid V^0) = \lambda A. \tag{93}$$

As before, the A found in (4) are invariant under the resistance scaling vector $\rho := \lambda^{-1}\widehat{\rho}$.

Example 13.4. See [Bar98, Example 6.7, p82]. *** Summarise***

13.5. **Abstract Existence Results.** To prove general existence results one needs to use a fixed point argument on a suitable space of Dirichlet forms in which scalar multiples of forms are identified.

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13.6. **Replication and Trace.** Suppose that $(\psi_1, \dots, \psi_M)$ is a pcf structure with base sets V^0 and V^1. Suppose (V^0, A) is a network and (V^1, A^R) is the replicated network using the resistance scaling vector $\rho = (\rho_1, \dots, \rho_M)$.

Suppose for $i = 1, \dots, M$ there is a network (W^i, B^i) such that $V^0 \subset W^i$, $\mathrm{Tr}(B^i \mid V^0) = A$, and $W^i \cap W^j = V^0$ for $i \neq j$.

For each i, one can apply (ψ_i, ρ_i) in the natural way to (W^i, B^i) and so obtain the replicated network (W^R, B^R). The trace of this network on V^1 is then (V^1, A^R).

More precisely, we have the following.

Proposition 13.5. *Suppose $F = (\psi_1, \dots, \psi_M)$ is a pcf structure with base sets V^0 and V^1. Suppose (V^0, A) is a network and $\rho = (\rho_1, \dots, \rho_M)$ is a resistance scaling vector. Let (V^1, A^R) be the replicated network.*

Suppose (W^i, B^i) is a network with conductance matrix $B^i = (b^i_{xy})$ for $1 \le i \le M$. Suppose $V^0 \subset W^i$, $\mathrm{Tr}(B^i \mid V^0) = A$, and $W^i \cap W^j = V^0$ for $i \neq j$.

Let (W^R, B^R) be the replicated matrix obtained by extending ψ_i to W^i so that $\psi_i(W^i) \cap \psi_j(W^j) = \psi_i(V^0) \cap \psi_j(V^0)$ for $i \neq j$, and by multiplying conductances in B^i by ρ_i analogous to (88). That is,

$$b^R_{xy} := \sum \{\rho_i b^i_{uv} : \psi_i(u) = x,\ \psi_i(v) = y\}. \tag{94}$$

Then the matrix B^R is a conductance matrix and the Dirichlet form $\mathcal{E}^R_B$ is given by

$$\mathcal{E}^R_B(f, g) := \frac{1}{2} \sum_{x,y} b^R_{xy} \big(f(x) - f(y)\big)\big(g(x) - g(y)\big) = \sum_{i=1}^{M} \rho_i \mathcal{E}_{B^i}(f \circ \psi_i, g \circ \psi_i). \tag{95}$$

Moreover,

$$\mathrm{Tr}\left(B^R \mid V^1\right) = A^R. \tag{96}$$

If A is invariant, then

$$\mathrm{Tr}\left(B^R \mid V^0\right) = A. \tag{97}$$

[26] *Note*: The map $A \mapsto \mathrm{Tr}(A^R \mid V^0)$ is nonlinear, but it *is* homogeneous. That is, if all conductances a_{xy} are multiplied by the same scalar α then $\mathrm{Tr}(A^R \mid V^0)$ is also multiplied by α. If A is invariant under ρ then so is any scalar multiple of A.

Proof. The fact $b^R_{xy} \geq 0$ if $x \neq y$ and the rows of B^R sum to zero follows from (94) and the analogous facts for the matrices B^i. The second equality in (95) similarly comes from summing over the sets W^i and the definitions.

For (96) it is sufficient to show that if $g : V^1 \to \mathbb{R}$ and f is its harmonic extension to (W^R, B^R) then

$$\mathcal{E}^R_A(g) = \mathcal{E}^R_B(f).$$

But f is the harmonic extension of g to (W^R, B^R) iff, for every i, $f \circ \psi_i$ is the harmonic extension of $g \circ \psi_i : V^0 \to \mathbb{R}$ to (W^i, B^i). To see this, consider $x \in W^R \backslash V^1$ and let i be the unique integer such that $x \in \psi_i(W^i)$, noting $x \notin V^1$ implies $x \notin \psi_i(V^0) \cap \psi_j(V^0) = \psi(W^i) \cap \psi_j(W^j)$ for $i \neq j$. Moreover, f is harmonic at x iff $f \circ \psi_f$ is harmonic at $\psi_f^{-1}(x)$, because all relevant conductances are scaled by the same factor ρ_i.

Finally

$$\begin{aligned} &\mathrm{Tr}(B^R \mid V^1) = A^R \quad \text{by (96)}, \\ &\mathrm{Tr}(A^R \mid V^0) = A \quad \text{since } A \text{ is invariant}, \\ \therefore\ &\mathrm{Tr}(B^R \mid V^0) = A \quad \text{by the tower property of traces (72)}. \end{aligned}$$

□

14. Sequences of Compatible Networks

*** to mainly follow [Bar98, 94–98], but using material already develoiped ***

15. Dirichlet Forms on PCF Fractals

*** to mainly follow [Bar05, 98–105] ***

15.1. Review of Dirichlet Forms and Laplacians.

A network $\mathbb{G} = (V, A)$ is given by a set V of *vertices* or *nodes* and a symmetric *conductance matrix* A such that

$$\begin{aligned} a_{xy} &= a_{yx} \geq 0 \quad \text{if } x \neq y, \\ a_{xx} &= -\sum_{\{y : y \neq x\}} a_{xy}. \end{aligned} \tag{98}$$

In particular, $\sum_y a_{xy} = 0$ for $x \in V$.

For $x \neq y$, $c_{xy} := a_{xy}$ is called the *conductance* of the edge xy. One defines $c_{xx} = 0$. The *conductance of the vertex* x is $c_x := \sum_{\{y:y \neq x\}} c_{xy} = -a_{xx}$.

We assume that the network is *connected* in the sense two vertices can be connected by a sequence of edges xy such that $a_{xy} > 0$.

There is a natural symmetric and irreducible Markov chain X defined by $P^x(X_1 = y) = c_{xy}/c_x$. In this setting c_x is the relative amount of time spend at x, and more precisely the unique steady state distribution is given by $c_x/\mathbb{C}(\mathbb{G})$ where $\mathbb{C}(\mathbb{G}) = \sum_x c_x$.

The *Dirichlet form* $\mathcal{E}$ is defined by

$$\begin{aligned} \mathcal{E}(f, g) &:= \frac{1}{2} \sum_{x,y} a_{xy} \big(f(x) - f(y)\big)\big(g(x) - g(y)\big) \\ &= -\sum_{x,y} f(x) a_{xy} g(y) = -f^T A g \quad \text{(bilinear form expression)} \\ &= -\sum_x f(x) \left(\sum_y a_{xy} g(y) \right) = -\sum_x f(x)\,(Ag)(x) \quad \text{(Greens's identity)}. \end{aligned} \tag{99}$$

The equivalence of these expressions is by simple algebra. The first expression is independent of the diagonal values a_{xx}. To obtain the second and third expressions use the fact $\sum_y a_{xy} = 0$ for each x.

The bracketed term in the third line is

$$
\begin{aligned}
Ag(x) &:= \sum_y a_{xy} g(y) = \sum_y a_{xy}\big(g(y) - g(x)\big) \\
&= c_x \sum_y \frac{c_{xy}}{c_x}\big(g(y)-g(x)\big) = c_x \left(\left(\sum_y \frac{c_{xy}}{c_x} g(y) \right) - g(x) \right) =: c_x\, \Delta^P g(x).
\end{aligned} \tag{100}
$$

We say $Ag(x)$ is the *natural Laplacian* of g at x and say $\Delta^P g(x)$ is the *probabilistic Laplacian* of g at x.

Note that $\Delta^P g(x)$ is the difference between a weighted average of values of g near x and $g(x)$ itself. It equals $\mathbb{E}^x g(X_1) - g(x)$. It also equals the total current into x if g represents voltage and current flows downhill.

If μ is a measure on V then from (99)

$$
\begin{gathered}
\mathcal{E}(f,g) = -(f, \Delta_\mu g)_\mu = -\sum_x f(x)\ \Delta_\mu g(x)\, \mu(x), \\
\text{where} \quad \Delta_\mu g(x) := \frac{1}{\mu(x)} Ag(x)
\end{gathered} \tag{101}
$$

is the μ-*Laplacian* and $(\cdot,\cdot)_\mu$ is the $L^2(\mu)$ inner product. That is, Δ_μ is defined so that Green's identity holds with the $L^2(\mu)$ inner product.

15.2. **Harmonic Extensions and Restrictions.** We mostly follow and extend [Str06, Section 1.3].

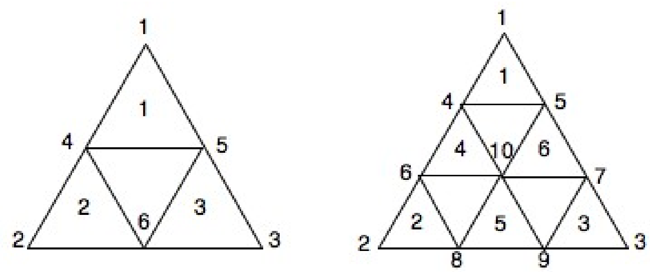

FIGURE 10. $SG(2)$ and $SG(3)$ prefractals with vertices and 1-cells numbered.

Consider a pcf structure $(K, \psi_1, \dots, \psi_M)$ with 0-cell V^0 having N vertices $(q_1, \dots, q_N)$.

15.2.1. *Mappings on* $\mathcal{H}$. The space $\mathcal{H}$ of functions defined on V^* (or $V^{(k)}$ or K), with prescribed values on V^0 and harmonic elsewhere, has dimension N. Identify $\mathcal{H}$ with $\mathbb{R}^N$. The basis vector $e_i = (0, \dots, 1, \dots 0)$, where the "1" occurs in the ith position, corresponds to the harmonic function f such that $f(q_j) = \delta_{ij}$.

The harmonic extension of $g : V^0 \to \mathbb{R}$ to $f : V^1 \to \mathbb{R}$ can be described in terms of linear transformations or matrices

$$
\begin{gathered}
M_i : \mathcal{H} \to \mathcal{H}, \quad M_i : \mathbb{R}^N \to \mathbb{R}^N, \quad i = 1, \dots M, \\
M_i(g) = f_i := f \circ \psi_i, \quad i = 1, \dots M.
\end{gathered} \tag{102}
$$

In effect, $M_i g$ is the restriction of f to K_i. Equivalently, $M_i g$ gives the values of f on the boundary vertices $\psi_i(q_j)$ of V^1, where the jth column gives the values in case $g = e_j$.

Example 15.1 ($SG(2)$). Using the $\frac{2}{5}, \frac{2}{5}, \frac{1}{5}$ rule for $g = e_1$ in Figure 10, $f(4) = f(5) = \frac{2}{5}$ and $f(6) = \frac{1}{5}$. Alternatively, the values of f at vertices $4, 5, 6$ are computed from $-A_{HH}^{-1} A_{HG} g$ as in (54). See (61) where G corresponds to the first three rows and columns, H to the last three, and A_{HH}^{-1} is the third matrix in (62).

This information gives the first column of the following 3 matrices. The other columns are similarly obtained by a symmetry argument.

$$M_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2/5 & 2/5 & 1/5 \\ 2/5 & 1/5 & 2/5 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 2/5 & 2/5 & 1/5 \\ 0 & 1 & 0 \\ 1/5 & 2/5 & 2/5 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 2/5 & 1/5 & 2/5 \\ 1/5 & 2/5 & 2/5 \\ 0 & 0 & 1 \end{bmatrix}. \tag{103}$$

Example 15.2 ($SG(3)$). The values of the harmonic extensions to V^1 of e_1, e_2, e_3 defined on V^0 in the right side diagram from Figure 10, are given by the three columns of

$$-A_{HH}^{-1}A_{HG} = \begin{bmatrix} \frac{8}{15} & \frac{4}{15} & 1/5 \\ \frac{8}{15} & 1/5 & \frac{4}{15} \\ \frac{4}{15} & \frac{8}{15} & 1/5 \\ \frac{4}{15} & 1/5 & \frac{8}{15} \\ 1/5 & \frac{8}{15} & \frac{4}{15} \\ 1/5 & \frac{4}{15} & \frac{8}{15} \\ 1/3 & 1/3 & 1/3 \end{bmatrix}.$$

The rows correspond to the vertices $4, \dots, 10$ in Figure 10. See (54) and (64), with G corresponding to the first three rows and columns and H to the last 7 rows and columns. Use Maple.

This gives

(104)

$$M_1 = \begin{bmatrix} 1 & 0 & 0 \\ 8/15 & 4/15 & 1/5 \\ 8/15 & 1/5 & 4/15 \end{bmatrix}, M_2 = \begin{bmatrix} 4/15 & 8/15 & 1/5 \\ 0 & 1 & 0 \\ 1/5 & 8/15 & 4/15 \end{bmatrix}, M_3 = \begin{bmatrix} 4/15 & 1/5 & 8/15 \\ 1/5 & 4/15 & 8/15 \\ 0 & 0 & 1 \end{bmatrix},$$
$$M_4 = \begin{bmatrix} 8/15 & 4/15 & 1/5 \\ 4/15 & 8/15 & 1/5 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}, M_5 = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/5 & 8/15 & 4/15 \\ 1/5 & 4/15 & 8/15 \end{bmatrix}, M_6 = \begin{bmatrix} 8/15 & 1/5 & 4/15 \\ 1/3 & 1/3 & 1/3 \\ 4/15 & 1/5 & 8/15 \end{bmatrix}.$$

15.2.2. *Mappings on $\widetilde{\mathcal{H}}$.* Note that $\mathcal{E}(u) = 0$ precisely on the constant functions. Factoring these out from $\mathcal{H}$ one obtains an $N-1$ dimensional Hilbert space $\widetilde{\mathcal{H}} = \mathcal{H}/\sim$ with the energy inner product

$$(f, g)_{\mathcal{E}} = \frac{1}{2} \sum_{x,y \in V^0} \big(f(x) - f(y)\big)\big(g(x) - g(y)\big).$$

A natural representative of the equivalence class $[g]$ is given by $g - (\overline{g}, \dots, \overline{g})$ where

$$\overline{g} = N^{-1} \sum_{x \in V^0} g(x), \quad \text{so} \quad \sum_{x \in V^0} \big(g(x) - \overline{g}\big) = 0.$$

It is convenient to work with a basis $\epsilon_1, \dots, \epsilon_{N-1}$ for $\widetilde{\mathcal{H}}$ which is orthonormal w.r.t. the energy inner product.

Example 15.3 ($SG(2)$). The vectors $(0,1,1)$ and $(0,1,-1)$ are $\mathcal{E}$ orthogonal and have energy norm $\frac{1}{2}(1+1) = 1$ and $\frac{1}{2}(1+1+2^2) = 3$ respectively. So we take the orthonormal basis

$$\epsilon_1 = (0,1,1) \text{ or } \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right), \quad \epsilon_2 = \left(0, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right),$$

where the second representative for ϵ_1 is the zero average representative.

It follows

$$M_1\epsilon_1 = \begin{bmatrix} 0 \\ 3/5 \\ 3/5 \end{bmatrix} = \frac{3}{5}\epsilon_1, \quad M_1\epsilon_2 = \begin{bmatrix} 0 \\ \sqrt{3}/15 \\ -\sqrt{3}/15 \end{bmatrix} = \frac{1}{5}\epsilon_2, \quad \therefore\ \widehat{M_1} = \begin{bmatrix} 3/5 & 0 \\ 0 & 1/5 \end{bmatrix};$$

$$M_2\epsilon_1 = \begin{bmatrix} 3/5 \\ 1 \\ 4/5 \end{bmatrix} \sim (0, 2/5, 1/5) = \frac{3}{10}\epsilon_1 + \frac{\sqrt{3}}{10}\epsilon_2,$$

$$M_2\epsilon_2 = \begin{bmatrix} \sqrt{3}/15 \\ \sqrt{3}/3 \\ 0 \end{bmatrix} \sim (0, 4\sqrt{3}/15, -\sqrt{3}/15) = \frac{\sqrt{3}}{10}\epsilon_1 + \frac{1}{2}\epsilon_2, \quad \therefore\ \widehat{M_2} = \begin{bmatrix} 3/10 & \sqrt{3}/10 \\ \sqrt{3}/10 & 1/2 \end{bmatrix};$$

$$M_3\epsilon_1 = \begin{bmatrix} 3/5 \\ 4/5 \\ 1 \end{bmatrix} \sim (0, 1/5, 2/5) = \frac{3}{10}\epsilon_1 - \frac{\sqrt{3}}{10}\epsilon_2,$$

$$M_3\epsilon_2 = \begin{bmatrix} -\sqrt{3}/15 \\ 0 \\ -\sqrt{3}/3 \end{bmatrix} \sim (0, \sqrt{3}/15, -4\sqrt{3}/15) = -\frac{\sqrt{3}}{10}\epsilon_1 + \frac{1}{2}\epsilon_2, \quad \therefore \widehat{M_3} = \begin{bmatrix} 3/10 & -\sqrt{3}/10 \\ -\sqrt{3}/10 & 1/2 \end{bmatrix}.$$

Example 15.4 ($SG(3)$). With the same energy orthonormal basis $\{\epsilon_1, \epsilon_2\}$ as in the previous example:

$$M_1\epsilon_1 = \begin{bmatrix} 0 \\ \frac{7}{15} \\ \frac{7}{15} \end{bmatrix} = \frac{7}{15}\epsilon_1, \quad M_1\epsilon_2 = \begin{bmatrix} 0 \\ \sqrt{3}/45 \\ -\sqrt{3}/45 \end{bmatrix} = \frac{1}{15}\epsilon_2, \quad \therefore\ \widehat{M_1} = \begin{bmatrix} 7/15 & 0 \\ 0 & 1/15 \end{bmatrix};$$

$$M_2\epsilon_1 = \begin{bmatrix} \frac{11}{15} \\ 1 \\ 4/5 \end{bmatrix} \sim (0, 4/15, 1/15) = \frac{1}{6}\epsilon_1 + \frac{\sqrt{3}}{10}\epsilon_2,$$

$$M_2\epsilon_2 = \begin{bmatrix} \sqrt{3}/9 \\ \sqrt{3}/3 \\ 4\sqrt{3}/45 \end{bmatrix} \sim (0, 2\sqrt{3}/9, -\sqrt{3}/45) = \frac{\sqrt{3}}{10}\epsilon_1 + \frac{11}{30}\epsilon_2, \quad \therefore\ \widehat{M_2} = \begin{bmatrix} 1/6 & \sqrt{3}/10 \\ \sqrt{3}/10 & 11/30 \end{bmatrix};$$

$$M_3\epsilon_1 = \begin{bmatrix} \frac{11}{15} \\ 4/5 \\ 1 \end{bmatrix} \sim (0, 1/15, 4/15) = \frac{1}{6}\epsilon_1 - \frac{\sqrt{3}}{10}\epsilon_2,$$

$$M_3\epsilon_2 = \begin{bmatrix} -\sqrt{3}/9 \\ -4\sqrt{3}/45 \\ -\sqrt{3}/3 \end{bmatrix} \sim (0, \sqrt{3}/45, -2\sqrt{3}/9) = -\frac{\sqrt{3}}{10}\epsilon_1 + \frac{11}{30}\epsilon_2, \quad \therefore \widehat{M_3} = \begin{bmatrix} 1/6 & -\sqrt{3}/10 \\ -\sqrt{3}/10 & 11/30 \end{bmatrix};$$

$$M_4\epsilon_1 = \begin{bmatrix} 7/15 \\ 11/15 \\ 2/3 \end{bmatrix} \sim (0, 4/15, 1/5) = \frac{7}{30}\epsilon_1 + \frac{\sqrt{3}}{30}\epsilon_2,$$

$$M_4\epsilon_2 = \begin{bmatrix} \sqrt{3}/45 \\ \sqrt{3}/9 \\ 0 \end{bmatrix} \sim (0, 4\sqrt{3}/45, -\sqrt{3}/45) = \frac{\sqrt{3}}{30}\epsilon_1 + \frac{1}{6}\epsilon_2, \quad \therefore\ \widehat{M_4} = \begin{bmatrix} 7/30 & \sqrt{3}/30 \\ \sqrt{3}/30 & 1/6 \end{bmatrix};$$

$$M_5\epsilon_1 = \begin{bmatrix} 2/3 \\ 4/5 \\ 4/5 \end{bmatrix} \sim (0, 2/15, 2/15) = \frac{2}{15}\epsilon_1, \quad M_5\epsilon_2 = \begin{bmatrix} 0 \\ 4\sqrt{3}/45 \\ -4\sqrt{3}/45 \end{bmatrix} = \frac{4}{9}\epsilon_2, \quad \therefore\ \widehat{M_5} = \begin{bmatrix} 2/15 & 0 \\ 0 & 4/9 \end{bmatrix};$$

$$M_6\epsilon_1 = \begin{bmatrix} 7/15 \\ 2/3 \\ 11/15 \end{bmatrix} \sim (0, 1/5, 4/15) = \frac{7}{30}\epsilon_1 - \frac{\sqrt{3}}{30}\epsilon_2,$$

$$M_6\epsilon_2 = \begin{bmatrix} -\sqrt{3}/45 \\ 0 \\ -\sqrt{3}/9 \end{bmatrix} \sim (0, \sqrt{3}/45, -4\sqrt{3}/45) = -\frac{\sqrt{3}}{30}\epsilon_1 + \frac{1}{6}\epsilon_2, \quad \therefore \widehat{M_6} = \begin{bmatrix} 7/30 & -\sqrt{3}/30 \\ -\sqrt{3}/30 & 1/6 \end{bmatrix}.$$

16. **PCF Fractals**

Here we follow [Str06, chapter 4].

16.1. **Basic Setup.** To simplify the notation, we deal with a restricted notion of *post crticially finite* [p.c.f.] fractals.

In the following assume K is invariant for the IFS $(F_1, \dots, F_N)$ in $\mathbb{R}^n$, where as usual the F_i are contractive. Then

$$K = \bigcup_{i=1}^{N} F_i K.$$

We are interested in the *connection* properties of K, and the contraction ratios do not play a role.

Definition 16.1 (PCF fractals)**.** K is *p.c.f.* if K is connected and there exists a finite $V_0 \subset K$ called the *boundary* such that

$$F_i K \cap F_j K \subset F_i V_0 \cap F_j V_o \quad \text{if } i \neq j, \tag{105}$$

and

$$V_0 \cap \bigcup_i F_i V_0 = \emptyset$$

$$V_0 = \{q_1, \dots, q_{N_0}\} \text{ for some } N_0 \leq N \text{ and } F_i q_i = q_i.$$

Remark 16.2.

(1) We can reorder the F_i if necessary to obtain $V_0 = \{q_1, \dots, q_{N_0}\}$.
(2) The requirement that "junction points" (see later) are fixed points is not necessary and is just to simplify the exposition. It rules out the Hata fractal.
(3) Equality clearly holds in (105).

□

Definition 16.3 (Approximating Graphs)**.**

$$V_m = \bigcup_i F_i V_{m-1} \text{ for } m \geq 1, \quad V_* = \bigcup_m V_m,$$

Γ_0 is the complete *graph* on V_0,

Γ_m is a *graph* on V_m such that:

$$\begin{aligned} x \sim_m y &\iff \exists i \big(x = F_i x'.\ y = F_i y',\ x' \sim_{m-1} y' \big) \\ &\iff \exists w \big(|w| = m,\ x, y \in F_w V_0 \big). \end{aligned}$$

Definition 16.4 (Cells and junction points)**.**

$F_w K$ for $|w| = m$ is an *m-cell*.

$F_w V_0$ is the *boundary* of $F_w K$.

Junction points are the intersection points of 2 m-cells, see (105).

Remark 16.5.

(1) V_m is the set of all boundary points of m-cells.
(2) The set of junction points at level m is a subset of V_m, but not necessarily conversely.
(3) A junction point x at level m is a junction point at level $m' > m$. The number of cells meeting at x is the same in each case.
(4) $K = \overline{V^*}$.
(5) Γ_m captures the connectivity properties of K at level m.

□

Example 16.6.

(1) $SG(k)$ for $k \geq 2$.
(2) Examples in [Str06, pages 101, 102]. These are the hexagasket, the hexagasket with smaller boundary, the Lindstrom snowflake, polygaskets and the Vicsek set.

16.2. **Energy.** We want a notion of energy $\mathcal{E}(u)$ for certain $u : V_* \to \mathbb{R}$, which is self-similar in the sense

$$\mathcal{E}(u) = \sum_{i=1}^{N} \rho_i \mathcal{E}(u \circ F_i) \tag{106}$$

for certain *resistance normalisation factors* $\rho_1, \ldots, \rho_N$ yet to be found.

For $SG(2)$ and $SG(3)$ it turns out that $\rho = 5/3$ and $15/7$ respectively.

16.2.1. *Construction of $\mathcal{E}$.* We aim to achieve this by approximating $\mathcal{E}$ by certain graph energies of the form

$$\mathcal{E}_m(u) = \sum_{x \sim_m y} c_m(x,y)\big(u(x) - u(y)\big)^2. \tag{107}$$

We extend this to the bilinear form

$$\mathcal{E}_m(u,v) = \sum_{x \sim_m y} c_m(x,y)\big(u(x) - u(y)\big)\big(v(x) - v(y)\big). \tag{108}$$

This leads to the *requirements*

$$\mathcal{E}(u) = \lim_{m \to \infty} \mathcal{E}_m(u), \tag{R1}$$

$$\mathcal{E}_m(u) = \sum_i \rho_i \mathcal{E}_{m-1}(u \circ F_i). \tag{R2}$$

To achieve (R2), note (R2) implies that once we define

$$\mathcal{E}_0(u) = \sum_{i<j} c_{ij}\big(u(q_i) - u(q_j)\big)^2, \quad c_{ij} > 0, \tag{109}$$

and fix the ρ_i, it follows that $\mathcal{E}_m(u)$ is determined by

$$\mathcal{E}_m(u) = \sum_{|w|=m} \sum_{i<j} \rho_w c_{ij}\big(u(F_w q_i) - u(F_w q_j)\big)^2. \tag{110}$$

Conversely, (110) implies (R2).

To achieve (R1) we require for $u : V_{m-1} \to \mathbb{R}$ and $m \geq 1$ that

$$\mathcal{E}_{m-1}(u) = \min\big\{\mathcal{E}_m(\widetilde{u}) \mid \widetilde{u} : V_m \to \mathbb{R},\ \widetilde{u} = u \text{ on } V_{m-1}\big\}. \tag{111}$$

Then

$$\mathcal{E}_m(u) \uparrow \ \text{ as } m \to \infty,$$

and so the limit on the right of (R1) exists, though it possibly equals ∞.

So now we only need to ensure by choice of the ρ_i and the c_{ij} that (111) is satisfied in case $m = 1$, since then it follows that (111) is satisfied for all m, essentially since the different m-cells behave independently of one another. Thus we require

$$\sum_{i<j} c_{ij}\big(u(q_i) - u(q_j)\big)^2 = \sum_k \sum_{i<j} \rho_k c_{ij}\big(\widetilde{u}(F_k q_i) - \widetilde{u}(F_k q_j)\big)^2, \tag{112}$$

where $\widetilde{u}$ minimises the right side subject to $\widetilde{u} = u$ on V_0.

The minimisation requirement, on differentiating w.r.t. $\widetilde{u}(x)$ for $x \in V_1 \setminus V_0$, gives $\widetilde{u}(x)$ as a linear combination of the $u(q_i)$ with coefficients determined by the ρ_k and c_{ij}. On substituting back into the RHS of (112) we obtain a nonnegative quadratic form

$$\widetilde{\mathcal{E}}_1(u) = \sum_{i<j} \widetilde{c}_{ij}\big(u(q_i) - u(q_j)\big)^2, \tag{113}$$

where the $\widetilde{c}_{ij}$ are function of the ρ_k and c_{ij}. Thus we require

$$\widetilde{c}_{ij} = c_{ij}. \tag{114}$$

(We can also treat this as a nonlinear eigenvalue problem and require $\lambda > 0$ such that

$$\widetilde{c}_{ij} = \lambda c_{ij}, \tag{115}$$

since in this case we can get (114) by replacing each ρ_k by $\lambda^{-1}\rho_k$.)

Henceforth we assume we do have (R1) *and* (R2), and hence from (110) that

$$c_m(x,y) = \rho_w c_{ij} \quad \text{if } |w| = m,\ x = F_w q_i,\ y = F_w q_j. \tag{116}$$

See [Str06, pp 104, 105] for discussion.

16.2.2. *Harmonic Extension Matrices.* Suppose that $\widetilde{u}$ is the unique minimiser of

$$\mathcal{E}_1(\widetilde{u}) = \sum_{x\sim_1 y} c_1(x,y)\big(\widetilde{u}(x) - \widetilde{u}(y)\big)^2,$$

subject to $\widetilde{u} = u$ on V_0. We call $\widetilde{u}$ the *harmonic extension* of u and say that u is *harmonic.*

Recall from (116) that $c_1(x,y)$ is given by $c_1(x,y) = \rho_k c_{ij}$ if $x = F_k q_i$ and $y = F_k q_j$.

Differentiating w.r.t. $\widetilde{u}(x)$ for $x \in V_1 \setminus V_0$ gives $\widetilde{u}(x)$ as a weighted sum of its neighbours, namely

$$\widetilde{u}(x) = \frac{\sum_{x\sim_1 y} c_1(x,y)\widetilde{u}(y)}{\sum_{x\sim_1 y} c_1(x,y)}.$$

This gives $\#(V_1 \setminus V_0)$ equations in $\#(V_1 \setminus V_0)$ unknowns. Since the minimiser exists the equations have a solution, and so the solution is unique. Thus $\widetilde{u}(x)$ for $x \in V_1 \setminus V_0$ is a linear combination of the $u(q_i)$. The averaging property implies a strict maximum principle and in particular $\widetilde{u}$ assumes its maximum and minimum on V_0. This also implies

$$\widetilde{u}(x) = \sum_{i=1}^{N_0} a_{xi} u(q_i) \text{ for } a_{xi} > 0.$$

***Why does Strichartz only claim ≥ 0 ?? ***

We may collect this information into the $N_0 \times N_0$ *harmonic extension matrices* A_i for $i = 1.\dots,N$ where

$$\begin{bmatrix} (\widetilde{u}\circ F_i)(q_1) \\ \vdots \\ (\widetilde{u}\circ F_i)(q_{N_0}) \end{bmatrix} = A_i \begin{bmatrix} u(q_1) \\ \vdots \\ u(q_{N_0}) \end{bmatrix}$$

The A_i are generally not invertible.

Substituting into (112) we obtain a formula for (113) and see that each $\widetilde{c}_{ij} > 0$. This implies from (115) that each $c_{ij} > 0$ and so we see why we needed the *complete* graph Γ_0 on V_0.

16.3. **Laplacians.** We assume that K is a p.c.f. fractal with $\mathcal{E}_0$ as in (109), $\mathcal{E}_m(u)$ as in (R2) for $\rho_i > 1$ or equivalently as in (110), and such that (111) is true.

The *domain* of $\mathcal{E}$ is

$$\mathcal{D}(\mathcal{E}) := \{i : \mathcal{E}(u) < \infty\}.$$

Let

$$\rho = \min \rho_i,\ r = \rho^{-1}\ (<1),\ r_w = \rho_w^{-1}.$$

Proposition 16.7. *If $x, y \in V_*$ belong to the same or adjacent m-cells then*

$$|u(x) - u(y)| \leq \frac{2r^{m/2}}{1 - r^{1/2}} \mathcal{E}(u)^{1/2}.$$

This implies

$$|u(x) - u(y)| \leq M|x - y|^{\gamma},$$

where M and γ depend on the Euclidean contraction constants for the F_i.

Proof. If $x \sim_m y$ then

$$|u(x) - u(y)| \leq r^{m/2}\mathcal{E}(u)^{1/2}.$$

This already gives continuity.

By a simple chaining argument, see [Str06, p28], if $x, y \in V_*$ belong to the same or adjacent m-cells, the first result follows.

The second follows by relating the contracting ratios of the F_i to r. □

Proposition 16.8. *$\mathcal{E}$ defines a Hilbert space on $\mathcal{D}(\mathcal{E})$ / constants.*

Proof. Straightforward. See [Str06, page 29]. □

Definition 16.9. We say that $u \in C(K)$ is *harmonic* if $u|_{V_m}$ is the harmonic extension of $u|_{V_{m-1}}$ for all $m \geq 1$. Note that u is completely determined by $u|_{V_0}$.

Proposition 16.10. *If $\widetilde{u}$ and $\widetilde{v}$ are the harmonic extensions $\widetilde{u}, \widetilde{v} : V_{m+1} \to \mathbb{R}$ of $u, v : V_m \to \mathbb{R}$ then*

$$\mathcal{E}_{m+1}(\widetilde{u}, \widetilde{v}) = \mathcal{E}_m(u, v).$$

In fact, if $v' : V_{m+1} \to \mathbb{R}$ is any exension of v then

$$\mathcal{E}_{m+1}(\widetilde{u}, v') = \mathcal{E}_m(u, v).$$

Proof. The first is by the polarisation identity. The second follows by considering $\mathcal{E}_{m+1}(\widetilde{u}, \widetilde{v} - v')$. See [Str06, Lemma 1.3.1, page 24]. □

Definition 16.11. The space $\mathcal{S}(\mathcal{H}_0, V_m)$ of *piecewise harmonic splines of level* m consists of those functions $u \in C(K)$ such that $u \circ F_w$ is harmonic for $|w| = m$. (See Section 16.2.2.)

For each $y \in V_m$, ψ_y^m is the piecewise harmonic m-spline such that $\psi_y^m(z) = \delta_y(z)$.

That is, $u \in \mathcal{S}(\mathcal{H}_0, V_m)$ is harmonic on K except on V_m. The function ψ_y^m is harmonic except on V_m, and equals 1 at y and 0 everywhere else on V_m.

Note that

(1) $u \in \mathcal{S}(\mathcal{H}_0, V_m)$ is characterised by $u|_{V_m}$. $\mathcal{S}(\mathcal{H}_0, V_m)$ has dimension $\#V_m$ with basis $\{\psi_y^m : y \in V_m\}$.
(2) $\mathcal{E}(u) = \mathcal{E}_m(u)$ if $u \in \mathcal{S}(\mathcal{H}_0, V_m)$.
(3) The ψ_y^m form a partition of unity on K. If $y \notin K_w$ then $\psi_y^m = 0$ on K_w.

Proposition 16.12. *$u \in C(K)$ is approximated uniformly by $u_m \in \mathcal{S}(\mathcal{H}_0, V_m)$ such that $u_m|_{V_m} = u|_{V_m}$.*

If $u \in \mathcal{D}(\mathcal{E})$ then $\mathcal{E}(u_m, u) \to 0$ as $m \to \infty$.

Proof. Straightforward. See [Str06, page 30]. □

The *effective resistance* $R(x, y)$ on K is defined as follows.

Definition 16.13. Fix $x, y \in V_m$. Then

$$R(x, y)^{-1} := \min\{\mathcal{E}(u) : u(x) = 0,\ u(y) = 1\}. \tag{117}$$

Equivalently, $R(x, y)$ is the least value of R such that

$$|u(x) - u(y)|^2 \leq R\,\mathcal{E}(u) \quad \forall u \in \mathcal{D}(\mathcal{E}).$$

For $x, y \in V_*$ it follows from the definition and the properties of harmonic minimisers that $R(x, y)$ is independent of m provided $x, y \in V_m$. This defines $R(x, y)$ for $x, y \in V_*$.

It is straightforward that $R(x, y)$ is continuous in x and y, and so by continuity is defined on K.

Moreover, it follows that (117) holds for $x, y \in K$.

Proposition 16.14. *R is a metric on K.*

Proof. It is sufficient to prove this on any finite network and then use continuity. This is standard. □

Proposition 16.15. *Let $x = F_w q_i$ and $y = F_w q_j$. Then*

$$\left(\sum_i c_{ij}\right)^{-1} r_w \leq R(x, y) \leq c_{ij}^{-1} r_w.$$

Suppose $x, y \in K$ belong to the same m-cell. Then

$$R(x, y) \leq c r_w.$$

Proof. Suppose ψ_y^m is the piecewise harmonic m-spline such that $\psi_y^m(z) = \delta_y(z)$. Then

$$R(x, y)^{-1} \leq \mathcal{E}(\psi_y^m) = \left(\sum_i c_{ij}\right) \rho_w.$$

Conversely, consider *any* u such that $u(x) = 0$ and $u(y) = 1$. Then

$$\mathcal{E}(u) \geq \mathcal{E}_m(u) \geq c_{ij} \rho_w,$$

and so $R(x,y)^{-1} \geq c_{ij}\rho_w$.

For the second result use a chaining argument. □

It follows that the resistance metric gives the Euclidean topology, but it is not normally equivalent to a power of the Euclidean metric.

We next give a definition of the "weak" μ-Laplacian w.r.t. a regular measure.

We use the notation $v \in \mathcal{D}_0(\mathcal{E})$ for $v \in \mathcal{D}(\mathcal{E})$ and $u = 0$ on V_0.

Definition 16.16. Suppose $u \in \mathcal{D}(\mathcal{E})$ and f is continuous. Then $u \in \mathcal{D}(\Delta_\mu)$ and $\Delta_\mu u = f$ if

$$\mathcal{E}(u,v) = -\int f v \, d\mu \quad \forall v \in \mathcal{D}_0(\mathcal{E}). \tag{118}$$

We use the notation

Definition 16.17. μ is a *self-similar measure* on K if $\exists \mu_i > 0$ such that

$$\mu(F_w K) = \prod_{i=1}^{N} \mu_{w_i} =: \mu_w.$$

It follows that μ has the scaling property

$$\int_K f \, d\mu = \sum_i \mu_i \int f \circ F_i \, d\mu \tag{119}$$

The Laplacian has the following scaling property.

Proposition 16.18. *If μ is a scaling measure then*

$$\Delta_\mu(u \circ F_w) = r_w \mu_w \, (\Delta_\mu u) \circ F_w. \tag{120}$$

Proof. Suppose $\Delta_\mu u = f$. From the definition (118) of the μ-Laplacian, using the scaling properties (106) and (120) in the left and right sides respectively,

$$\sum_{i=1}^{N} \rho_i \mathcal{E}(u \circ F_i, v \circ F_i) = -\sum_i \mu_i \int_K (f \circ F_i)(v \circ F_i) \, d\mu,$$

for $v \in \mathcal{D}_0(\mathcal{E})$.

Fix i and $w \in \mathcal{D}_0(\mathcal{E})$. Let $v \circ F_i = w$ and $v \circ F_j = 0$ if $j \neq i$. Then it follows that

$$\rho_i \mathcal{E}(u \circ F_i, w) = -\mu_i \int_K (f \circ F_i) \, w \, d\mu.$$

From (120) it follows

$$\rho_i \, \Delta_\mu(u \circ F_i) = \mu_i \, (\Delta_\mu u) \circ F_i.$$

Iterating gives the required result. □

Informally, we may hope that harmonic functions u as in Definition 16.9 are characterised by $\Delta_\mu u = 0$, essentially since the measures correspond to a conformal factor at each vertex. This is the case, and more is true.

We introduce the pointwise Laplacians.

Definition 16.19. The *graph (or discrete) Laplacian* Δ_m on Γ_m (based on the conductances) is

$$\Delta_m u(x) = \sum_{y \sim_m x} c_m(x,y)\big(u(y) - u(x)\big). \tag{121}$$

In the following theorem one can think of the right side of (122) as a graph laplacian on V_m in which the edges xy are weighted by $c_m(x,y)$ and the vertices x are weighted by $\left(\int_K \psi_x^m d\mu\right)^{-1}$. This *modified* graph laplacian approaches the weak laplacian without any renormalisation factors.

Note that by the remarks preceding Proposition 16.12, if μ is a self-similar measure then the the denominator in the following is simple to evaluate. Namely

$$\int_K \psi_x^m d\mu = \frac{1}{3} \sum \{\mu_w : |w| = m, \ x \in f_w V_0\}.$$

This assumes V_0 is a triangle, otherwise change the 3.

Theorem 16.20. *Suppose $u \in \mathcal{D}(\Delta_\mu)$. Then*

$$\Delta_\mu u(x) = \lim_{m\to\infty} \frac{\Delta_m u(x)}{\int_K \psi_x^m d\mu}. \tag{122}$$

and the convergence is uniform on $V_* \setminus V_0$.

Suppose $u \in C(K)$ and the right side of (122) converges uniformly on $V_* \setminus V_0$. Then $u \in \mathcal{D}(\Delta_\mu)$ and (122) holds.

Proof. See [Str06, Theorem 2.2.1, page 44].

The idea is to first suppose $x \in V_m \setminus V_0$ for m large.

Let $\Delta_\mu = f$. Then

$$\mathcal{E}(u,v) = -\int_K fv\, d\mu \quad \forall v \in \mathcal{D}_0(\mathcal{E}).$$

Let $v = \psi_x^m$. Then

$$\mathcal{E}(u,\psi_x^m) = \mathcal{E}_m(u,\psi_x^m) = -\Delta_m u(x),$$

where the first inequality comes from Proposition 16.10 and the second from the definitions of $\Delta_m u$ and ψ_x^m. Moreover,

$$\int_K f\psi_x^m d\mu \approx f(x)\int_K \psi_x^m d\mu = \Delta_\mu u(x)\int_K \psi_x^m d\mu.$$

Hence

$$\Delta_\mu(x) \approx \frac{\Delta_m u(x)}{\int_K \psi_x^m d\mu}.$$

□

Thus we can regard (122) as a pointwise definition of the Laplacian $\Delta_\mu u(x)$.

Definition 16.21. At a boundary point $q_i \in V_0$ the *normal derivative* of u is defined by

$$\partial_n u(q_i) = \lim_{m\to\infty} \rho_i^m \sum_{j\neq i} c_{ij}\big(u(q_i) - u(F_i^m q_j)\big) = -\lim_{m\to\infty} \Delta_m u(q_i),$$

provided the limit exists.

Note the sequence is constant if u is harmonic. *** Write out the argument ***

For the following, c.f. [Str06, Theorem 2.3.2., page 47].

Theorem 16.22 (Gauss-Green theorem)**.** *Suppose $u \in \mathcal{D}(\Delta_\mu)$. Then $\partial_n u(q_i)$ exists for all $q_i \in V_0$. For all $v \in \mathcal{D}(\mathcal{E})$ the following Gauss-Green formula holds:*

$$\mathcal{E}(u,v) = -\int_K (\Delta_\mu u)v\, d\mu + \sum_{q_i\in V_0} v(q_i)\partial_n u(q_i). \tag{123}$$

Proof. First take $v(q_i) = 1$ and $v(q_j) = 0$ if $j \neq i$ in (118). □

Remark 16.23. Hence

(1) $\int_K \Delta_\mu u\, d\mu = \sum_{q_i\in V_0} \partial_n u(q_i)$.
(2) u harmonic $\implies \sum_{q_i\in V_0} \partial_n u(q_i) = 0$.

□

*** Following Corollary to be checked***

Corollary 16.24. *If $u \in \mathcal{S}(\mathcal{H}_0, V_m)$ and $v \in \mathcal{D}(\mathcal{E})$ then*

$$\mathcal{E}(u,v) = \mathcal{E}_m(u,v) = \sum_{|w|=m}\sum_{p\in\partial V_w} v(p)\partial_n^{F_wV_0} u(p).$$

Moreover, for $p = q_i$ and $w = i\cdots i$,

$$\partial_n^{F_wV_0} u(p) := \lim_{n\to\infty} -\Delta_n u(p) = -\Delta_m u(p).$$

A similar result applies to every F_wV_0.

Proof. The first equality is from Proposition 16.10 but going from V_m directly to V_M for $M > m$ rather than from V_m to V_{m+1}.

Equality between the first and third quantities is from Theorem 16.22 applied to each $F_w V_0$ separately.

For the last assertion, the sequence is constant for $n \geq m$ by the harmonicity of u in $F_w V_0$, and we have just taken the mth member of the sequence. This requires that the appropriate e-value of the harmoni extension matrix is the renormalisation constant. See [Str06]: page 26, last paragraph before exercises; page 48, Figure 2.3.1 □

*** Summarise the material on Greens function, on pages 116–119 of [Str06]. ***

*** Summarise the material on non self-similar fractals on on pages 122–124 of [Str06]. ***

17. Hieararchical Fractals

We consider a fractal built as a limit of scaled copies of the $SG(2)$ and $SG(3)$ prefractals, see Figure 11.

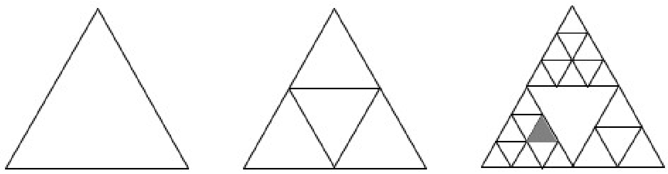

Figure 11. Prefractals for a hierarchical gasket; Γ_0, Γ_1 and Γ_2 from left to right.

Remark 17.1 (Basic set up).
1. Γ_0 is the complete graph on V_0, which has 3 vertices.
2. $V_0 \subset V_1 \subset V_2 \subset \cdots$, $V_* = \bigcup_m V_m$.
3. V_* is dense in K.
4. V_m and K are partitioned into overlapping m-cells, each with 3 boundary points in V_m.
5. $x \sim_m y$ if x and y are in a common m-cell.
6. If C is an m-cell then the way the parent $m-1$ cell is subdivided is given by $n(C)$, where $n(C) = 2, 3$ according as C is part of a scaled copy of $SG(2)$ or $SG(3)$ respectively.
   Thus if C is the top 1-cell in V_1 from Figure 11 then $n(C) = 2$; if it is the shaded cell then $n(C) = 3$.

□

Definition 17.2. K as constructed above is a *hierarchical gasket.* If $n(C)$ depends only on the level of C then K is a *homogeneous hierarchical gasket.*

Remark 17.3 (Words and other Notation).
1. Consider words $w = w_1 w_2 w_3 \ldots$ which may be finite or countably infinite in length. Here $w_i \in \{1, 2, 3\}$ or $\{1, 2, \ldots, 6\}$.
2. Cells at level m are denoted C_w with length $|w| = m$. The cell at level 0 is denoted $C_\emptyset$.
   The shaded 2-cell in Figure 11 is C_{14}, using the convention of numbering the maps corresponding to $SG(2)$ and $SG(3)$ in a counterclockwise manner beginning from the bottom left "corner".
3. $w_m \in \{1, 2, 3\}$ or $\{1, 2, \ldots, 6\}$ according as $n(C_{w_1 \ldots w_m}) = 2$ or 3 respectively.

□

Remark 17.4 (Renormalisation Factor). The cell m-cell C_w is given the *conductance renormalisation factor* $\rho_w = \rho_{w_1} \cdot \ldots \cdot \rho_{w_m}$ where $\rho_{w_i} = 5/3$ or $15/7$ according as $n(C_{w_1 \ldots w_i}) = 2$ or 3 respectively.

That is, one multiplies by $5/3$ or $15/7$ according as the cell belongs to a scaled copy of $SG(2)$ or $SG(3)$ respectively. Corresponding to the shaded cell C_{14}, $\rho_{14} = \frac{5}{3} \times \frac{15}{7}$.

Thus

$$\rho_w = \left(\frac{5}{3}\right)^{m_2} \times \left(\frac{15}{7}\right)^{m_3}.$$

where m_2 is the number of occurrences of 2 among $n(C_{w_1\dots w_i})$ for $1 \le i \le m$, and m_3 is the number of occurrences of 3.

The *resistance renormalisation factor* is $r_w := \rho_w^{-1}$. □

Remark 17.5 (Energy). The *energy functional* on V_m is defined by

$$\mathcal{E}_m(u) = \sum_{|w|=m} \sum_{x \sim_m y} \rho_w \big(u(x) - u(y)\big)^2.$$

It follows

$$\mathcal{E}_m(\widetilde{u}) = \mathcal{E}_{m-1}(u) \tag{124}$$

where $\widetilde{u}$ is the harmonic extension to V_m of $u : V_{m-1} \to \mathbb{R}$.

Moreover,

$$\mathcal{E}_m(u) \uparrow \text{ as } m \to \infty, \quad \mathcal{E}(u) := \lim_{m\to\infty} \mathcal{E}_m(u) \tag{125}$$

exists (possibly as ∞) for $u : V_* \to \mathbb{R}$ and hence similarly for $u \in C(K)$. □

Remark 17.6. The properties of energy and harmonic functions in Section 16.3 all hold, except for the scaling properties.

In particular, the diameter of a cell C_w in the resistance metric is comparable to r_w. □

Remark 17.7 (Laplacian). Let μ be the standard measure with

$$\mu(C_w) = \mu_w = \left(\frac{1}{3}\right)^{m_2} \times \left(\frac{1}{6}\right)^{m_3},$$

although any other measure is similar.

The *discrete Laplacian* for $x \in V_m$ is

$$\Delta_m u(x) = \sum_{y \sim_m x} c_m(x, y)\big(u(x) - u(y)\big),$$

where $c_m(x, y) = \rho_w$ if $x, y \in C_w$, c.f. Definition 16.19.

The *weak Laplacian* is

$$\Delta_\mu u(x) = \lim_{m\to\infty} \frac{\Delta_m u(x)}{\int_K \psi_x^m \, d\mu},$$

by the same proof as in Theorem 16.20.

It is straightforward to show that for $x \in V_m$,

$$\int_K \psi_x^m \, d\mu = \sum_j \frac{1}{3} \mu_{w^j}$$

where C_{w^j} are the m-cells containing x, of which there are 1, 2 or 3. If $x \in V_m \setminus V_{m-1}$ then all the μ_{w^j} are equal. □

*** Include the material on Greens function on page 124 of [Str06]. Also Prop 1.9 and Theorem 1.12 of [?KL]. ***

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